Turbo screening

Primal-dual screening tests for the LASSO problem

September 29, 2020

Theo Guyard
theo.guyard@insa-rennes.fr
INSA Rennes, 4GM

Cedric Herzet
cedric.herzet@inria.fr
INRIA Rennes, SIMSMART

Clement Elvira
clement.elvira@inria.fr
INRIA Rennes, PANAMA

Abstract

This document is a technical support resulting from two months of a joint project between a student at INSA Rennes and two researchers at INRIA. It aims to explain the path that led to the development of new screening tests for the LASSO problem, allowing to improve the convergence speed of many solving methods. Screening tests already existed to tackle the LASSO when the dimension was too large to be handled. They are based on geometrical considerations in its dual problem. They aim to reduce iteratively the dimension during the iterations of a solving method. We have developed new screening tests that can significantly improve the convergence and that can be used with many algorithms. They allow to bypass the last iterations where methods are usually slow in getting closer to the solution. These new tests are based on geometrical considerations in the primal problem. To efficiently solve the LASSO, we overload a solving method with screening tests in both primal and dual problems. We called this approach Turbo screening. In particular, we apply this method to FISTA, ADMM and the Frank-Wolfe algorithm. In some setups, we manage to divide by more than 100 the number of operations needed to met the target precision.

Keywords – LASSO, screening, primal screening, dual screening, gap region, ellipsoidal region, FISTA, ADMM, Frank-Wolfe, Active-set.

Contents

1 Introduction **3**
- 1.1 Non-negative LASSO . . . 3
- 1.2 Problem derivations . . . 4
- 1.3 Reduced problem . . . 5
- 1.4 Outline of the method . . . 6

2 Screening tests **9**
- 2.1 Primal screening tests . . . 9
  - 2.1.1 General primal screening test . . . 9
  - 2.1.2 Some primal screening regions . . . 9
- 2.2 Dual screening tests . . . 10
  - 2.2.1 General dual screening test . . . 10
  - 2.2.2 Some dual screening regions . . . 11
- 2.3 Turbo screening in practice . . . 12
  - 2.3.1 Heuristic dual gap screening . . . 13
  - 2.3.2 Safeness behaviour of the relaxed dual gap sphere . . . 15

3 Turbo screening algorithm **17**
- 3.1 Generic algorithm . . . 17
- 3.2 Implementation details . . . 18

4 Numerical results **19**
- 4.1 Experimental setup . . . 19
- 4.2 Convergence scheme in extreme cases . . . 20
  - 4.2.1 FISTA . . . 20
  - 4.2.2 ADMM . . . 21
  - 4.2.3 Frank-Wolfe . . . 21
- 4.3 Heuristic dual screening performance . . . 22

5 Conclusion **23**

Appendix A Numerical results **24**
- A.1 FISTA . . . 24
- A.2 ADMM . . . 25
- A.3 FW . . . 26

Appendix B Optimization methods **27**
- B.1 FISTA . . . 27
  - B.1.1 Basic step . . . 27
  - B.1.2 Backtracking . . . 27
  - B.1.3 Taking advantage of FISTA . . . 28
- B.2 ADMM . . . 28
  - B.2.1 Problem and basic step . . . 28
  - B.2.2 Application to the non-negative LASSO . . . 30
  - B.2.3 Taking advantage of ADMM . . . 30

B.3 Frank-Wolfe 30
B.3.1 Basic step 30
B.3.2 Improvements 31
B.3.3 Application to non-negative LASSO 31
B.4 Active-set method 31
B.4.1 General problem and theory 32
B.4.2 Application to non-negative LASSO 33

Appendix C Proofs **34**
C.1 Proof of (2.2.2) 34
C.2 Proof of (2.2.4) 34
C.3 Proof of (2.2.3) 34
C.4 Proof of (2.3.1) 34

1 Introduction

1.1 Non-negative LASSO

The LASSO problem is a very famous problem which finds applications in Statistics, Signal Processing, Machine Learning and many other fields. It aims to find a sparse linear approximation of some data. For a given *data vector* $\mathbf{y} \in \mathbb{R}^m$ and a given *dictionary* $\mathbf{A} \in \mathbb{R}^{m \times n}$, it seeks for a sparse vector (*i.e.* with lot of null entries) $\mathbf{x}^\star \in \mathbb{R}^n$ such that $\mathbf{y} \simeq \mathbf{A}\mathbf{x}^\star$. When we need $\mathbf{x}^\star \geq \mathbf{0}_n$, this problem is rather called the non-negative LASSO problem. It can be expressed as

$$\mathbf{x}^\star \in \underset{\mathbf{x} \geq \mathbf{0}_n}{\operatorname{argmin}} \left\{ \mathrm{P}(\mathbf{x}) = \tfrac{1}{2} \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \lambda \mathbf{1}_n^{\mathrm{T}} \mathbf{x} \right\} \tag{$\mathcal{P}$}$$

where $\lambda \geq 0$ is called the *regularization parameter*. The classical LASSO problem can always be transformed into a non-negative LASSO problem. In the following, we abusively refer to ($\mathcal{P}$) as the LASSO problem. The first term of P ensures that $\mathbf{A}\mathbf{x}^\star$ well approximates $\mathbf{y}$ and the second term penalizes non-sparse solutions. The parameter λ acts as a trade-off : the larger λ, the sparser $\mathbf{x}^\star$ but the larger $\|\mathbf{y} - \mathbf{A}\mathbf{x}^\star\|$. For any $\lambda \geq 0$,

$$\operatorname{card}(\operatorname{supp}(\mathbf{x}^\star)) \leq m \tag{1.1.1}$$

where $\operatorname{supp}(\mathbf{x}^\star) = \{i, \mathbf{x}_i^\star > 0\}$. Thus, the solution obtained by solving ($\mathcal{P}$) is enough sparse. The value $\lambda_{\max} := \max(\mathbf{A}^{\mathrm{T}}\mathbf{y})$ is the highest regularization value. For $\lambda \geq \lambda_{\max}$, the solution of ($\mathcal{P}$) is $\mathbf{x}^\star = \mathbf{0}_n$. When $\lambda = 0$, ($\mathcal{P}$) is just a least-square problem. The ratio $0 \leq \lambda/\lambda_{\max} \leq 1$ is often used to tune *regularization amount* targeted in $\mathbf{x}^\star$.

As ($\mathcal{P}$) is a convex problem, many methods allow to tackle it. Gradient-based methods are often used. The most famous one is FISTA [BT09] which has already shown some very good performances. Other algorithm such as ADMM [BPC11] or the Frank-Wolfe algorithm [Jag13] also allow to deal with this problem. When a machine-epsilon accuracy is needed, Active-set method [NW06] can also be used. However, some setups can give these algorithms a hard time. When the dimension is too large, computations with $\mathbf{A}$ or even a memory load of this matrix can be too costly to handle. Furthermore, when $\mathbf{A}$ has highly correlated columns and when the regularization is low, algorithms usually struggle to converge when a high accuracy is required.

Recently, screening tests [GVR10][XWR16] have been developed to face dimensionality issues. They allow to reduce the dimension of $\mathbf{A}$ and $\mathbf{x}$ during the iterations of an algorithm. These tests are based on geometrical considerations in the dual problem of ($\mathcal{P}$) and aim to identify the saturated positivity constraints in $\mathbf{x}^\star$. Hereafter, we call them *primal screening tests*. In this report, we present new screening tests, constructed similarly to those already developed but based on geometrical considerations directly in ($\mathcal{P}$). We called these tests *dual screening tests*. With these tests, we manage to identify the support of $\mathbf{x}^\star$ (non-saturated constraints of ($\mathcal{P}$)) which allows to simply solve a small auxiliary linear system to recover $\mathbf{x}^\star$. In this way, we manage to stop the algorithm much sooner and bypass

the last iterations where methods are usually slow. Using both primal and dual screening tests alongside, we manage to handle tricky cases when $\mathbf{A}$ has highly correlated columns of when a small regularization is set and to recover the solution with a machine epsilon precision. We called this approach *Turbo screening*.

1.2 Problem derivations

Thereafter, we assume that each column $\mathbf{a}_i$ of $\mathbf{A}$ and $\mathbf{y}$ are ℓ_2-normalized. In the following, we use the dual problem of ($\mathcal{P}$) which can be parametrized as

$$\mathbf{u}^\star = \underset{\mathbf{u} \mid \mathbf{A}^{\mathrm{T}}\mathbf{u} \leq \mathbf{1}}{\operatorname{argmax}} \left\{ \mathrm{D}(\mathbf{u}) = \tfrac{1}{2}\|\mathbf{y}\|_2^2 - \tfrac{\lambda^2}{2}\left\|\tfrac{\mathbf{y}}{\lambda} - \mathbf{u}\right\|_2^2 \right\} \tag{$\mathcal{D}$}$$

One can notice that the feasible region of ($\mathcal{D}$) is a polytope defined by the intersection $\mathcal{U} = \cap_{i=0}^{n} \mathcal{U}_i$ where $\mathcal{U}_i = \{\mathbf{u}, \mathbf{a}_i^{\mathrm{T}}\mathbf{u} \leq 1\}$. By definition, ($\mathcal{D}$) have a unique maximizer $\mathbf{u}^\star := \mathrm{proj}_{\mathcal{U}}(\mathbf{y}/\lambda)$. Even though its expression is simple, the dual optimal variable $\mathbf{u}^\star$ is as hard to compute as the primal optimal variable $\mathbf{x}^\star$.

In the following, $\mathbf{x}_{\mathrm{f}}$ and $\mathbf{u}_{\mathrm{f}}$ denotes respectively a primal and a dual feasible points. Using this notation, we can define the duality gap as

$$\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) = \mathrm{P}(\mathbf{x}_{\mathrm{f}}) - \mathrm{D}(\mathbf{u}_{\mathrm{f}}) \geq 0$$

Given a primal-dual pair $(\mathbf{x}, \mathbf{u})$, we can construct a primal-dual feasible pair $(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$ using the following *primal scaling* and *dual scaling* operations :

$$\mathbf{x}_{\mathrm{f}} = \max\{\mathbf{0}\,; \mathbf{x}\} \tag{1.2.1a}$$

$$\mathbf{u}_f = \begin{cases} \mathbf{u} & \text{if } \max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{u} \leq 1 \\ \frac{\mathbf{u}}{\max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{u}} & \text{otherwise} \end{cases} \tag{1.2.1b}$$

The first operation is just a projection onto the positive orthant while the second one consists in sliding $\mathbf{u}$ toward the origin until is reaches $\mathcal{U}$.

To design screening tests, we need two of the optimality conditions of ($\mathcal{P}$) which are

$$\mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star \in \begin{cases} \{1\} & \text{if } \mathbf{x}_i^\star > 0 \\ ]-\infty, 1] & \text{if } \mathbf{x}_i^\star = 0 \end{cases} \tag{1.2.2a}$$

$$\mathbf{A}\mathbf{x}^\star = \mathbf{y} - \lambda\mathbf{u}^\star \tag{1.2.2b}$$

The first condition informs about which constraints are saturated in ($\mathcal{P}$) and in ($\mathcal{D}$) whereas the second one links $\mathbf{x}^\star$ and $\mathbf{u}^\star$. In ($\mathcal{P}$) and ($\mathcal{D}$), we can define the *primal-active set* and the *dual-active set* as

$$\mathcal{I}_{\mathrm{p}}^\star = \{i \in [\![1, n]\!],\ \mathbf{x}_i^\star = 0\} \qquad \text{and} \qquad \mathcal{I}_{\mathrm{d}}^\star = \{i \in [\![1, n]\!],\ \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star = 1\}$$

In the following, we assume that $\mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star = \emptyset$ so $\mathcal{I}_{\mathrm{p}}^\star$ and $\mathcal{I}_{\mathrm{d}}^\star$ are complementary. In this case, one has $\mathcal{I}_{\mathrm{d}}^\star = \mathrm{supp}(\mathbf{x}^\star)$. Thus, we can deduce from (1.2.2a) that

$$\mathbf{x}_i^\star > 0 \iff \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star = 1 \tag{1.2.3a}$$

$$\mathbf{x}_i^\star = 0 \iff \mathbf{a}_i^{\mathrm{T}}\mathbf{u}^\star \in\,]-\infty, 1[\tag{1.2.3b}$$

In practice, $\mathbf{A}$ is sampled randomly.In such cases, one has $\mathbb{P}(\mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star \neq \emptyset) = 0$ regardless of the generation law used if it is continuous. We discuss how our method behaves without this hypothesis in the next paragraph.

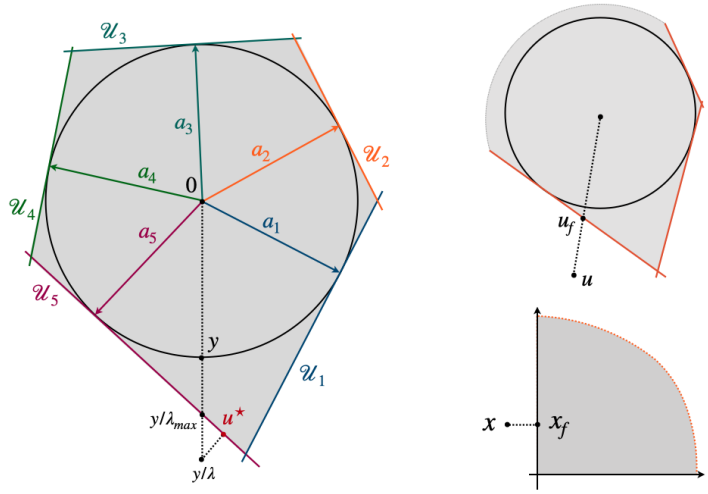

Figure 1: Left : Dual LASSO geometry. Right : primal-dual scaling operations.

1.3 Reduced problem

Assume that $\mathcal{I}_{\mathrm{p}}^\star$ (or equivalently $\mathcal{I}_{\mathrm{d}}^\star$) is known. Let $\tilde{\mathbf{A}} = [\mathbf{a}_i]_{i \in \mathcal{I}_{\mathrm{d}}^\star}$ and let $\tilde{\mathbf{x}} = [\mathbf{x}_i]_{i \in \mathcal{I}_{\mathrm{d}}^\star}$. One has from (1.2.2a) and (1.2.2b) that

$$\tilde{\mathbf{A}}^{\mathrm{T}} \tilde{\mathbf{A}} \tilde{\mathbf{x}}^\star = \tilde{\mathbf{A}}^{\mathrm{T}} \mathbf{y} - \lambda \mathbf{1} \tag{1.3.1}$$

The dimension of (1.3.1) is relatively small in the classical use-cases of ($\mathcal{P}$) because in practice card$(\mathcal{I}_{\mathrm{d}}^\star) << m$. We can notice that contrary to (1.2.2b), $\mathbf{u}^\star$ is not involved. The solution of ($\mathcal{P}$) can be reconstructed from $\tilde{\mathbf{x}}^\star$ as

$$\begin{cases} \mathbf{x}_i^\star = \tilde{\mathbf{x}}_i^\star & \text{if } i \in \mathcal{I}_{\mathrm{d}}^\star \\ \mathbf{x}_i^\star = 0 & \text{if } i \in \mathcal{I}_{\mathrm{p}}^\star \end{cases} \tag{1.3.2}$$

Thus, knowing $\mathcal{I}_{\mathrm{p}}^\star$ or $\mathcal{I}_{\mathrm{d}}^\star$ allows to solve directly (1.3.1) and use (1.3.2) to recover the solution of ($\mathcal{P}$). As a linear system can be solved with a high accuracy, $\mathbf{x}^\star$ can be recovered with a machine epsilon precision. However we do not have any prior information about $\mathcal{I}_{\mathrm{p}}^\star$ or $\mathcal{I}_{\mathrm{d}}^\star$ in general.

We will see that using primal and dual screening tests, we can identify respectively elements of $\mathcal{I}_{\mathrm{p}}^\star$ and $\mathcal{I}_{\mathrm{d}}^\star$. During the iterations of a solving method, these tests are proceeded and when all the elements of $[\![1, n]\!]$ have been *primal screened* or *dual screened* (*i.e.* identified as belonging to $\mathcal{I}_{\mathrm{p}}^\star$ or $\mathcal{I}_{\mathrm{d}}^\star$), we can solve (1.3.1) and use (1.3.2) to recover $\mathbf{x}^\star$ with a high accuracy.

We assumed above that $\mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star = \emptyset$. Actually, we will see that primal and dual screening tests can only identify elements of $\overline{\mathcal{I}_{\mathrm{p}}^\star}$ and $\overline{\mathcal{I}_{\mathrm{d}}^\star}$. With (1.2.2a), we can deduce that $\overline{\mathcal{I}_{\mathrm{d}}^\star} \subset \mathcal{I}_{\mathrm{p}}^\star$ and $\overline{\mathcal{I}_{\mathrm{p}}^\star} \subset \mathcal{I}_{\mathrm{d}}^\star$. Thus, screening tests can identify correctly a subset of the primal and dual active sets. However, if $i_0 \in \mathcal{I}_{\mathrm{p}}^\star \cap \mathcal{I}_{\mathrm{d}}^\star$, then neither primal nor dual screening tests can identify i_0 as belonging to $\mathcal{I}_{\mathrm{p}}^\star$ or to $\mathcal{I}_{\mathrm{d}}^\star$. The consequence is that $\mathcal{I}_{\mathrm{p}}^\star$ and $\mathcal{I}_{\mathrm{d}}^\star$ can't be fully identified so it is impossible to switch

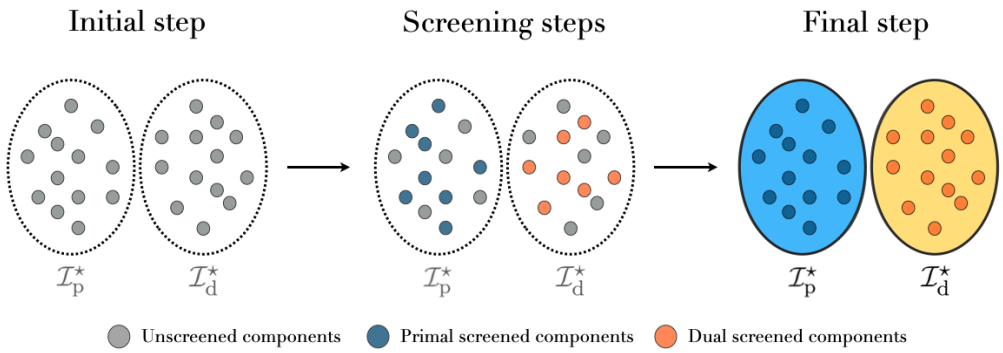

Figure 2: Illustration of iterative primal and dual screening tests identifying if elements belong to $\mathcal{I}_p^\star$ or $\mathcal{I}_d^\star$. At the final step, one can switch to the resolution of (1.3.1). Each item is an elements of $[\![1, n]\!]$.

to the resolution of (1.3.1). As we mentioned above, such unfavourable cases do not often occur in practice. Furthermore, we propose an heuristic in 2.3 working even if $\mathcal{I}_p^\star \cap \mathcal{I}_d^\star \neq \emptyset$.

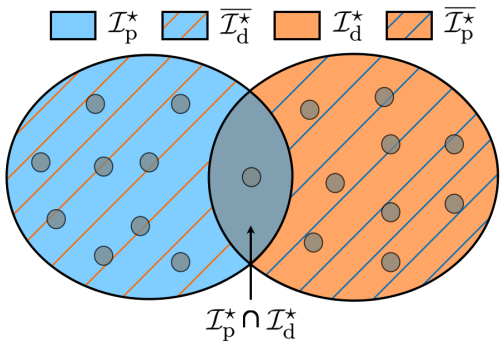

Figure 3: An unfavourable case where $\mathcal{I}_p^\star \cap \mathcal{I}_d^\star \neq \emptyset$. Screening tests can only identify components in hatched regions.

1.4 Outline of the method

The aim of Turbo Screening is to improve a method in term of number of operations required to met a target precision on the duality gap. Only multiplications are counted as additions have a marginal cost in practice. It consists in these successive steps within the iterations of the chosen solving method :

- Compute a new iterate with the method
- Identify components of $\mathcal{I}_p^\star$ using primal screening tests
- Identify components of $\mathcal{I}_d^\star$ using dual screening tests
- Solve (1.3.1) and use (1.3.2) to get $\mathbf{x}^\star$ if $\mathcal{I}_p^\star$ and $\mathcal{I}_d^\star$ are fully identified

Primal screening tests already presented in literature allow to reduce the dimension of the problem. For any i identified as belonging to $\mathcal{I}_p^\star$, one can remove

the i-th column of $\mathbf{A}$ and the i-th coefficient of $\mathbf{x}$ during the resolution process, making the remaining problem simpler and faster to solve. Some primal screening tests allow to identify entirely $\mathcal{I}_{\mathrm{p}}^{\star}$ under a certain dual gap when the method is converging toward the solution. At this point, the dimension cannot be reduced any more. Once $\mathcal{I}_{\mathrm{p}}^{\star}$ is fully identified, we can simply solve the linear system (1.3.1) to recover the solution of ($\mathcal{P}$) with a very high accuracy.

The downside is that we do not know when the identification of $\mathcal{I}_{\mathrm{p}}^{\star}$ is completed. No theoretical results ensure that the dual gap is enough small so that $\mathcal{I}_{\mathrm{p}}^{\star}$ is entirely screened. This motivates us to design new dual screening tests allowing to identify components of $\mathcal{I}_{\mathrm{d}}^{\star}$ in a similar way. We can use primal and dual screening tests alongside to monitor whether $\mathcal{I}_{\mathrm{p}}^{\star}$ has been fully identified or not. When every components of $[\![1, n]\!]$ are identified as belonging to $\mathcal{I}_{\mathrm{p}}^{\star}$ or $\mathcal{I}_{\mathrm{d}}^{\star}$, we are sure that these sets are fully known. Instead of running the method until the stopping criterion is met, we rather switch to the resolution of (1.3.1) when the identification of $\mathcal{I}_{\mathrm{p}}^{\star}$ and $\mathcal{I}_{\mathrm{d}}^{\star}$ is completed in order to recover the solution of ($\mathcal{P}$) with a very high accuracy. We called this approach *Turbo screening* as it speeds up the resolution process. We will see that an heuristic similar to a greedy active set strategy [WN99] can be used with dual screening tests to speed up even more the convergence speed.

The be efficient, it is advised that the solving method used with Turbo screening has the following properties :

Pre-computed values To avoid a too large additional computational cost brought by the screening tests, we can re-use values already computed by the method. Actually, we only need the value of $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$ to proceed screening tests. One can obtain this value in $o(n+m)$ if the value $\nabla \mathrm{P}$ is already computed. As $\nabla \mathrm{P}$ is computed in $o(nm)$, the additional computational cost brought by screening tests is marginal. Gradient-based methods compute $\nabla \mathrm{P}$ in their iterations so they are particularly well suited for Turbo screening but many other methods also use this value within their iterations.

Duality gap Screening tests used in Turbo screening are even more efficient when when dual gap is tight. If $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \to 0$ in the method used, $\mathcal{I}_{\mathrm{p}}^{\star}$ and $\mathcal{I}_{\mathrm{d}}^{\star}$ can be fully identified below a certain dual gap. In a method providing iterative points, one can use (1.2.2a), (1.2.1a) and (1.2.1b) to construct a feasible pair $(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$. Thus, new points provided by the method will strengthen the screening tests and will allow to identify more and more components of $\mathcal{I}_{\mathrm{p}}^{\star}$ and $\mathcal{I}_{\mathrm{d}}^{\star}$ through the iterations.

We will see that some of our dual screening tests already compute the values needed to solve the final linear system (1.3.1), making it cheap in computations to solve too.

In section 2, we first review primal screening tests, we present our new dual

screening tests and some heuristics. In section 3, we explain our final method and some numerical results are presented in sections 4. Methods usually used to solve ($\mathcal{P}$) and implemented for our numerical results are detailed in appendix B.

2 Screening tests

Screening tests were first introduced in [GVR10] for the LASSO problem. They initially aimed to reduce the dimension of the problem by identifying null entries of $\mathbf{x}^\star$ before the resolution process. Later, dynamic screening tests [Bon+14] were developed in the aim of using FISTA iterations to create stronger screening tests during the algorithm. A very efficient test is the gap screening test [Ndi+17] which allows to identify all the null entries of $\mathbf{x}^\star$ if points provided by FISTA are enough close to $\mathbf{x}^\star$. The key of screening tests is to use values already computed by FISTA or by another optimization method in order to be proceed at a very low computational cost. In the following, screening tests already introduced in literature are called *primal screening tests* as they allow to identify $\mathcal{I}_{\mathrm{p}}^\star$. These tests are reviewed in the first subsection and then, our new *dual screening tests* allowing to identity $\mathcal{I}_{\mathrm{d}}^\star$ are presented.

2.1 Primal screening tests

The aim of primal screening tests is to *screen* (*i.e.* identify) elements of $\mathcal{I}_{\mathrm{p}}^\star$. When i is identified as belonging to $\mathcal{I}_{\mathrm{p}}^\star$, one can remove the i-th coefficient and column of $\mathbf{x}$ and $\mathbf{A}$ in order to reduce the dimension of the problem and to make the solving method to run faster.

2.1.1 General primal screening test

Primal screening tests are based on the following geometrical intuition. Let a region $\mathcal{R}$ in the dual space such that $\mathbf{u}^\star \in \mathcal{R}$. Using (1.2.2a), one can derive that

$$\max_{\mathbf{u}\in\mathcal{R}} \mathbf{u}^{\mathrm{T}}\mathbf{a}_i < 1 \Longrightarrow i \in \mathcal{I}_{\mathrm{p}}^\star \tag{2.1.1}$$

This test is proceeded for each coefficient i of $\mathbf{x}$ (or equivalently each column of $\mathbf{A}$). In the literature, several types of region were proposed. The most famous are spherical and dome regions [XWR16]. As we do not want to penalize too much the solving method used, we want $\mathcal{R}$ to make (2.1.1) easy to solve. That is why, simple regions are often used. We also want $\mathcal{R}$ to have a high screening rate. The more coefficients $\mathcal{R}$ can screen, the more the dimension can be shrunk. If $\mathcal{R}_1 \subset \mathcal{R}_2$, then any coefficient screened by $\mathcal{R}_2$ is also screened by $\mathcal{R}_1$. Thus, we always seek for a tight region bounding $\mathbf{u}^\star$ to proceed primal screening tests. As for most optimizations methods, the dual gap tends to 0, regions with a size decreasing with the dual gap are particularly suited to proceed primal screening tests. They give the certificate that $\mathcal{I}_{\mathrm{p}}^\star$ can be fully identified below a certain dual gap.

2.1.2 Some primal screening regions

Spherical regions allow to solve (2.1.1) easily and are often used. With $\mathcal{R} \equiv \mathcal{S}(\mathbf{c}, r)$, the solution of (2.1.1) is $\max_{\mathbf{u}\in\mathcal{S}(\mathbf{c},r)} \mathbf{a}_i^{\mathrm{T}}\mathbf{u} = \mathbf{a}_i^{\mathrm{T}}\mathbf{c} + r$.

ST1 region Using properties of the projection operator, one can derive that

$$\|\mathbf{u}^\star - \mathbf{y}/\lambda\|_2 \leq 1/\lambda - 1/\lambda_{max} \tag{2.1.2}$$

Thus, we can use the spherical region $\mathcal{R} \equiv \mathcal{S}(\mathbf{y}/\lambda, 1/\lambda - 1\lambda_{\max})$ to proceed a primal screening test. Although this region doesn't have a high screening rate and is used for static screening (*i.e.* region is not strengthen at each iteration of the optimization method), it is a good example to understand how screening works. Improved ST2 and ST3 spherical regions can also be created in the same way [XWR16].

Primal GAP region Given a primal and a dual feasible pair $(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})$ and using the Lipschitz property of $\nabla \mathrm{P}$ in ($\mathcal{P}$), one can derive [Ndi+17] that

$$\|\mathbf{u}^\star - \mathbf{u}_\mathrm{f}\|_2 \leq \sqrt{2\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})}/\lambda \tag{2.1.3}$$

Thus, one can use the spherical region $\mathcal{R} \equiv \mathcal{S}(\mathbf{u}_\mathrm{f}, \sqrt{2\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})}/\lambda)$ to proceed a primal screening test. This region is very interesting because one can use the points constructed by the solving method along with (1.2.2b) and (1.2.1b) to construct the pair $(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})$ so as to proceed primal screening tests. All the values needed to proceed the primal gap test are cheap to get if $\nabla \mathrm{P}$ is already known. Furthermore, the region is tightened at each iteration as $\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f}) \to 0$ and below a certain gap, all the elements of $\mathcal{I}_\mathrm{p}^\star$ will have been identified.

More sophisticated regions Regions consisting of an intersection between a spherical region and an hyperplane are called dome regions [XWR16]. As they are tighter, they have a better screening rate than spherical regions. However, they are often more costly to construct but (2.1.1) is still easy to solve with these regions. Using this principle, it is also possible to construct spherical regions intersected with several hyperplanes. Screening regions can also be constructed using non-expansiveness property of the projection operator [Wan+13] [MH16]. Joint screening tests were also developed to test more than one coefficient with only one screening test [HDD19].

2.2 Dual screening tests

In the following, we present new dual screening tests allowing to identify components of $\mathcal{I}_\mathrm{d}^\star$. They are constructed in a same way as primal screening tests.

2.2.1 General dual screening test

Dual screening tests construction arises from the following geometrical intuition. Let a region $\mathcal{R}$ in the primal space such that $\mathbf{x}^\star \in \mathcal{R}$, then

$$\min_{\mathbf{x} \in \mathcal{R}} x_i > 0 \Longrightarrow i \in \mathcal{I}_\mathrm{d}^\star \tag{2.2.1}$$

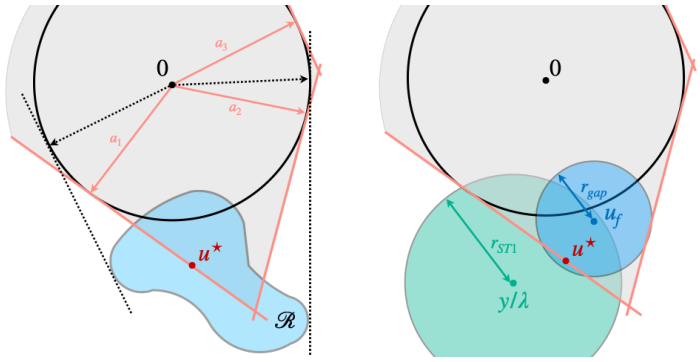

Figure 4: Left : Geometrical representation of primal screening tests. Here, all the index i such that $\mathbf{a}_i$ is outside the two dotted thresholds (such as $a3$) are screened by the blue region. Right : ST1 and primal gap region.

Likewise in primal screening tests, the challenge is to find a good region $\mathcal{R}$ in the sense that it can dual screen as many components as possible and it allows to solve (2.2.1) easily. Typically, the smaller the region, the better the screening rate. Regions with a simple shape allow usually to solve (2.2.1) easily. For example, considering a spherical region $\mathcal{R} \equiv \mathcal{S}(\mathbf{c}, r)$ leads to $\min_{\mathbf{x}\in\mathcal{S}(\mathbf{c},r)} x_i = c_i - r$. Some algorithms such as the Ellipsoid method [BGT81] already construct regions bounding $\mathbf{x}^\star$ during their resolution process. However, fastest methods known to solve ($\mathcal{P}$) do not directly give such regions.

2.2.2 Some dual screening regions

In the following, we propose new regions to proceed dual screening tests with a good screening rate and making (2.2.1) easy to solve.

Dual ST1 region The simplest dual screening region can be constructed similarly to the primal ST1 sphere. Given a primal feasible point $\mathbf{x}_\mathrm{f}$, one has

$$\|\mathbf{x}^\star - \mathbf{0}_n\|_2 \leq \mathrm{P}(\mathbf{x}_\mathrm{f})/\lambda \tag{2.2.2}$$

Thus, one can solve (2.2.1) with $\mathcal{R} \equiv \mathcal{S}(\mathbf{0}_n, \mathrm{P}(\mathbf{x}_\mathrm{f})/\lambda)$. However, likewise the primal ST1 sphere, this region doesn't have a high screening rate. We don't use it in our final Turbo screening method but this region is a good example to understand how dual screening works.

Dual gap region A good dual screening region can be constructed similarly to the primal gap region. If $n \leq m$ and given $\mathbf{x}_\mathrm{f}$ and $\mathbf{u}_\mathrm{f}$, one has

$$\|\mathbf{x}^\star - \mathbf{x}_\mathrm{f}\|_2 \leq \sqrt{2\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})}/\alpha \tag{2.2.3}$$

for any $0 < \alpha \leq \lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$ where $\lambda_{\min}(\mathbf{M})$ is the smallest eigenvalue of $\mathbf{M}$. The finest region for dual screening is created with $\alpha = \lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$. One can use the spherical dual spherical region $\mathcal{R} \equiv \mathcal{S}(\mathbf{x}_\mathrm{f}, \sqrt{2\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f})}/\alpha)$. Assuming that a value $0 < \alpha \leq \lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$ is known, a dual screening test with this region has a complexity in $o(1)$ in order to be proceeded.

Remark 2.2.1. *In an algorithm using Turbo-screening, one can first find a feasible value for α and use it for the rest of the algorithm, even if the dimension of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ is shrunk by primal screening tests. Indeed, $\lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ is non-decreasing when columns of $\mathbf{A}$ are removed [Hwa04]. Using the first value computed for α allows to create dual gap regions for the rest of the algorithm but these regions will not be optimal in the sense that smaller dual gap spheres could have been used.*

Dual Ellipsoidal region Ellipsoidal regions also allow to solve (2.2.1) with cheap computations. Any ellipsoid region can be parametrized using a covariance matrix $\mathbf{E}$ and a center $\mathbf{c}$ as $\mathcal{E}(\mathbf{E}^{-1}, \mathbf{c}) = \{\mathbf{x}, (\mathbf{x}-\mathbf{c})^{\mathrm{T}}\mathbf{E}^{-1}(\mathbf{x}-\mathbf{c}) \leq 1\}$. Given a pair $(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$, one can use the following inequality to bound $\mathbf{x}^\star$ inside an ellipsoidal region :

$$(\mathbf{x}-\mathbf{x}_{\mathrm{f}})^{\mathrm{T}}\mathbf{E}^{-1}(\mathbf{x}-\mathbf{x}_{\mathrm{f}}) \leq 1 \tag{2.2.4}$$

where $\mathbf{E}^{-1} = \mathbf{A}^{\mathrm{T}}\mathbf{A}/2\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$. One needs $n \leq m$ otherwise the region created is degenerated. This region can be strengthened by taking the intersection with the hyperplane

$$\mathcal{H}(\mathbf{g}, \mathbf{x}_{\mathrm{f}}) = \{\mathbf{x}, \mathbf{g}^{\mathrm{T}}(\mathbf{x}-\mathbf{x}_{\mathrm{f}}) \leq 0\} \tag{2.2.5}$$

where $\mathbf{g} = \nabla(\mathrm{P}\mathbf{x}_{\mathrm{f}})$. This intersection makes the region twice smaller than the initial ellipsoid region $\mathcal{E}(\mathbf{E}^{-1}, \mathbf{x}_{\mathrm{f}})$. The solutions of (2.2.1) with $\mathcal{R} \equiv \mathcal{E}(\mathbf{E}^{-1}, \mathbf{x}_{\mathrm{f}})$ or with $\mathcal{R} \equiv \mathcal{E}(\mathbf{E}^{-1}, \mathbf{x}_{\mathrm{f}}) \cap \mathcal{H}(\mathbf{g}, \mathbf{x}_{\mathrm{f}})$ are given in [MdM19]. The region $\mathcal{E}(\mathbf{E}^{-1}, \mathbf{x}_{\mathrm{f}}) \cap \mathcal{H}(\mathbf{g}, \mathbf{x}_{\mathrm{f}})$ has a better screening rate but the complexity to solve (2.2.1) is $o(n^2)$ whereas the simpler region $\mathcal{E}(\mathbf{E}^{-1}, \mathbf{x}_{\mathrm{f}})$ has a worst (but still good) screening rate and the complexity to solve (2.2.1) is only $o(n)$. The complexity given above takes into account that $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ is known. An advantage of using a dual ellipsoidal screening test is that it allows to solve (1.3.1) easily as $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ is already computed. Once the initial $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ is computed, we can use rank-one rules to update it every time columns of $\mathbf{A}$ are removed by primal screening tests. The updated inverse can be retrieved in $o(n^2)$ using the following remark.

Remark 2.2.2. *Assume that $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ is known. One want to compute the updated inverse with a column deletion in $\mathbf{A}$ (last column without loss of generality). Let $\mathbf{B} = (\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ and let $\tilde{\mathbf{B}}$ the updated inverse. Then, $\tilde{\mathbf{B}} = \mathbf{B}_{11} - d\mathbf{u}_2\mathbf{u}_2^{\mathrm{T}}$ where*

$$\begin{cases} \mathbf{B}_{11} &= \mathbf{B}(1:n-1, 1:n-1) \\ d &= \mathbf{B}(n,n) \\ \mathbf{u}_3 &= -\mathbf{B}(1:n-1, n) \\ \mathbf{u}_2 &= \mathbf{u}_3/d \end{cases} \tag{2.2.6}$$

The dual gap region and the dual ellipsoidal region have both a size decreasing with the dual gap. Thus, using them in an algorithm in which $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \to 0$ allows to identify entirely $\mathcal{I}_{\mathrm{d}}^\star$ below a certain dual gap.

2.3 Turbo screening in practice

Primal screening tests using the gap region (2.1.3) are really cheap to proceed when $\nabla\mathrm{P}$ can be extracted from the optimization method iterations and allow to

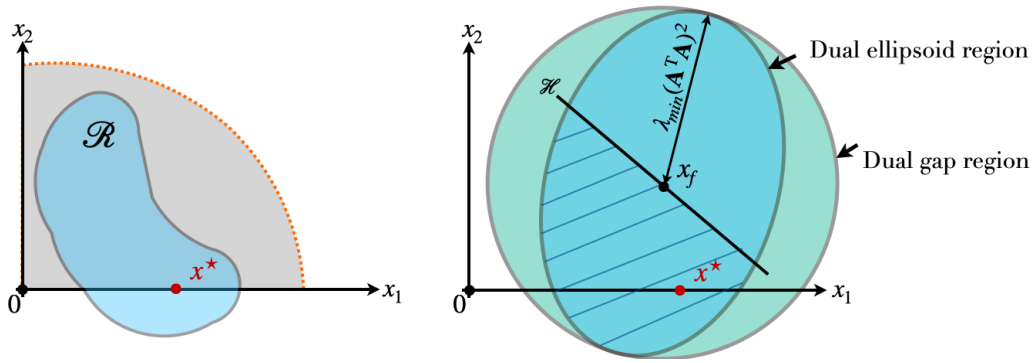

Figure 5: Left : A dual screening region alloing to deduce that $1 \in \mathcal{I}_{\mathrm{d}}^\star$. Right : Dual gap and ellipsoid regions with the hyperplane $\mathcal{H}$ used to strengthen the ellipsoid region. In this case, we can conclude that $1 \in \mathcal{I}_{\mathrm{d}}^\star$ with any of the three regions.

reduce the problem complexity. Dual screening tests do not allow to reduce the problem complexity. Furthermore, they are cheap to proceed as long as one know a suitable value of α in (2.2.3) or the value of $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ in (2.2.4). However, such values are not known in most of the methods used to solve ($\mathcal{P}$). Computing these values in order to proceed dual screening tests may over-penalize the method, leading ultimately to more computations with Turbo screening than without. Of course, if the method used allows to get α or $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ cheaply, Turbo screening can be used as it. In the following, we propose two heuristics to use Turbo screening efficiently in practice when α and $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ are unknown.

2.3.1 Heuristic dual gap screening

A suitable value for α in (2.2.3) is usually hard to compute. In some methods such as FISTA, one only have informations about $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. The idea of the heuristic is to relax the radius of the dual gap sphere by setting a value $\alpha \geq \lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. A region for which the associated screening test can not be mistaken is called 'safe'. Thus, the region created in this way is theoretically unsafe. However, numerical experiments have shown that if the value chosen for α is not too far from $\lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ and if the current point is not too far from $\mathbf{x}^\star$, the dual screening test proceeded with the relaxed dual gap sphere is very rarely wrong when identifying components of $\mathcal{I}_{\mathrm{d}}^\star$. Even more, most of the time, the region allows to identify entirely $\mathcal{I}_{\mathrm{d}}^\star$ in only one dual screening test.

In algorithm such as FISTA, one can set $\alpha = L$ where L is the Lipschitz constant used (*c.f.* (B.1.2a)). If $\mathbf{A}$ is not ill-conditioned, one can hope to create a region which is not too unsafe. When no values such as L are known, it is possible to set $\alpha = \min(\|\mathbf{A}\|_1, \|\mathbf{A}\|_\infty)$. A third option is to use Gershgorin theorem [Bel65] to find an upper bound on $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. If $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ is diagonal dominant, this theorem also provides a strictly positive lower bound on $\lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ which can be used to generate a safe dual gap sphere. When the matrix $\mathbf{A}$ is sampled randomly, the probability that $\mathbf{A}^{\mathrm{T}}\mathbf{A}$ is diagonal dominant increases with its dimension. There are many other ways to find a suitable value for α in order to created a relaxed radius.

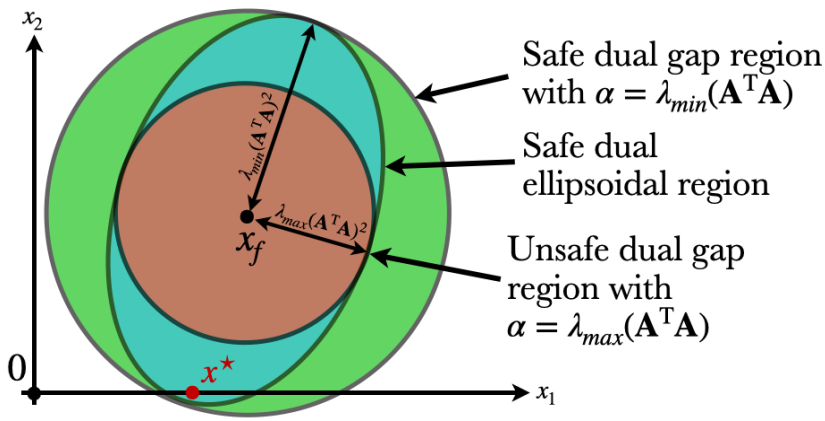

Figure 6: Safe and unsafe dual regions. Green and yellow regions safely identify $1 \in \mathcal{I}_{\mathrm{d}}^\star$. Red region also unsafely identify $2 \in \mathcal{I}_{\mathrm{d}}^\star$.

Using the relaxed dual gap sphere, one can proceed screening tests with a very low additional computational cost. However, some mistakes can still be made during the dual screening part due to the unsafeness of the region. That is why one must continue to proceed primal screening tests on elements assumed to belong to $\mathcal{I}_{\mathrm{d}}^\star$ in order to correct possible mistakes. Two strategies can be used to be sure that the solution obtained with the heuristic dual gap screening is actually the solution of ($\mathcal{P}$).

Mistakes correction strategy : From the top of the algorithm, proceed primal screening tests. Once $n \leq m$, enable unsafe dual gap test. When all the components have been safely primal screened or unsafely dual screened, one can hope that only few mistakes were made and that the primal and dual active set are almost well identified. As card($\mathcal{I}_{\mathrm{d}}^\star$) is very small in practice, the dimension of the remaining problem is enough small to compute $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ with a reasonable computation complexity. Using this value, one can proceed safe ellipsoidal screening test to correct dual screening mistakes and use rank-one update rules [Kha08] to update the matrix inverse without the columns wrongly dual screened. One can then solve (1.3.1) and use (1.3.2) to get $\mathbf{x}^\star$. If only few mistakes were made by the relaxed dual gap test, few safe dual screening test are needed to correct all the mistakes.

Greedy Active-set strategy : From the top of the algorithm, proceed primal screening tests. Once $n \leq m$, enable unsafe dual gap test. Every time that new components are heuristically dual screened, solve the linear system (1.3.1) using the current know dual active set. The value of $(\tilde{\mathbf{A}}^{\mathrm{T}}\tilde{\mathbf{A}})^{-1}$ is computed the first time that an element of $\mathcal{I}_{\mathrm{d}}^\star$ is identified and then, rank-one update rules can be used every time the knowledge of $\mathcal{I}_{\mathrm{d}}^\star$ is updated. Every time that (1.3.1) is solved, we can use an optimality criterion (for example the duality gap fo the reduced problem) to check is $\mathcal{I}_{\mathrm{d}}^\star$ is well identified or not. There are three advantages with this heuristic :

- $\mathrm{card}(\mathcal{I}_{\mathrm{d}}^{\star})$ is usually low so the first computation of $(\tilde{\mathbf{A}}^{\mathrm{T}}\tilde{\mathbf{A}})^{-1}$, rank-one update rules and the resolution of (1.3.1) can be made with a reasonable computational cost
- We do not have to wait that $\mathcal{I}_{\mathrm{p}}^{\star}$ is fully identified by primal screening tests. If the solution of (1.3.1) satisfies the chosen optimality criterion, we can stop the algorithm and use (1.3.2) to recover $\mathbf{x}^{\star}$
- Even if $\mathcal{I}_{\mathrm{p}}^{\star} \cap \mathcal{I}_{\mathrm{d}}^{\star} \neq \emptyset$ as assumed in 1.3, this heuristic can identify entirely $\mathcal{I}_{\mathrm{d}}^{\star}$ as the radius of the region is relaxed

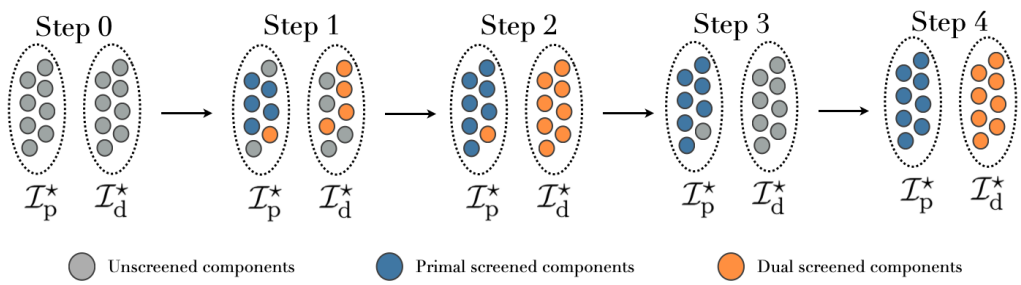

Figure 7: Relax and correct mistakes strategy. Step 0 : Initialization. Step 1 : Safe primal screening and unsafe dual screening. Step 2 : All the coefficients have been screened but with possible mistakes. Step 3 : Safe dual screening tests are triggered at a low dimension, dual screened coefficients are checked. Step 4 : All the coefficients have been safely screened. The reduced linear system is solved and the solution is recover with a machine-epsilon accuracy.

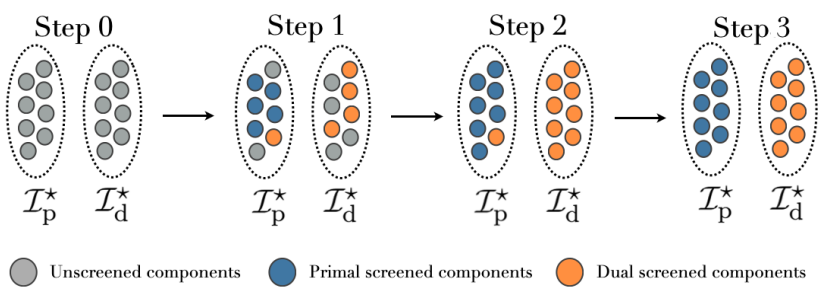

Figure 8: Greedy Active-set strategy. Step 0 : Initialization. Step 1 : Safe primal screening and unsafe dual screening. Step 2 : All the coefficients have been screened but with possible mistakes. Reduced linear system is solved but the solution is not optimal yet. Step 3 : Safe primal screening allows to correct unsafe dual screening mistakes. Step 4 : All the coefficients have been screened without mistakes. The solution of the reduced linear system known as optimal using optimality criteria of the reduced problem. The solution of ($\mathcal{P}$) is reconstructed with a machine-epsilon accuracy.

2.3.2 Safeness behaviour of the relaxed dual gap sphere

The following theorem gives a mathematical criterion on the safeness of the dual gap sphere with a relaxed radius.

Theorem 2.3.1. *Let $\kappa(\mathbf{M})$ the condition number of $\mathbf{M}$ and let $\tilde{\alpha} > 0$ a lower bound on $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. For any $c \geq \kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A})$, the region created using $\alpha = \tilde{\alpha}/c$ is safe.*

In practice, we don't know $\kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ but we can tune the safety of the region with the parameter c. Even if c is very small (making the radius small and the region very unsafe), the region doesn't produce lots of mistakes. An intuition is that primal screening tests already remove 'tricky' coefficients which could have been dual screened wrongly. If the implementation is made such that primal screening tests are proceeded before dual screening tests, most of the tricky coefficients are removed by primal screening tests before that the unsafe dual screening test can produce mistakes. Furthermore, in most practical cases, one initially has $m \leq n$. Thus, some iterations are already needed before that dual screening tests can be proceeded. When $n \leq m$ and that dual screening tests are enabled, the current point is usually already close from $\mathbf{x}^\star$. Thus, the duality gap is enough small to make the dual gap region very unlikely to produce mistakes.

Example 2.3.1. *The initial point used is* $\mathbf{x}^0 = \mathbf{0}$. *If the first iteration of FISTA gives a point* $\mathbf{x}^1$ *as in the following graphic.*

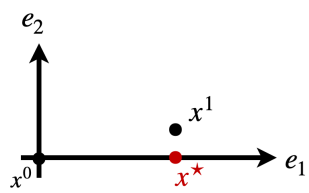

Figure 9: A favourable case where the unsafe dual gap region screens safely.

Here, the coefficient along e_2 *can possibly be dual screened wrongly as* $x_2^1 > 0$ *but* $x_2^\star = 0$. *As primal screening tests are proceeded before unsafe dual screening tests in the implementation (but also because primal screening tests begin at iteration 0 whereas dual screening tests with the relaxed dual gap region can only begin once* $m \leq n$*), the coefficient along* e_2 *will probably be primal screened very soon in the algorithm, making any dual screening mistakes unlikely to occur. If dual screening tests were implemented before primal screening tests and if* $m \leq n$*, the configuration below may have lead to a mistake by dual screening the coefficient* $x_2^\star$ *if* α *is chosen very small.*

3 Turbo screening algorithm

Now that all the theoretical background was developed, one can go through some practical details in order to implement an efficient Turbo screening algorithm. The aim is to derive a generic way to use this approach, working for almost any optimization method allowing to solve ($\mathcal{P}$).

3.1 Generic algorithm

The following algorithm is generic and can be applied to many optimization methods allowing to solve ($\mathcal{P}$). We choose to use the greedy active-set heuristic explained in section 2.3. Primal screening tests are proceeded using the primal gap sphere (2.1.3). Dual screening tests are proceeded using the dual gap sphere (2.2.3) with a relaxed radius. One can notice that the values of $\mathbf{Ax}$ and of $\mathbf{A}^{\mathrm{T}}\mathbf{u}$ where $\mathbf{u} = \mathbf{y} - \mathbf{Ax}$ are outputted by a method iteration. These values are used to proceed screening tests with a low computational cost. For most methods, these values can be extracted as they are involved in the computation of $\nabla \mathrm{P}$. However, if it is not possible to extract these values, they can still be computed but the computational cost of the screening tests will be more important. Primal screening tests are always proceeded before dual screening tests in order to remove tricky coefficients which could have been dual screening wrongly by the relaxed dual gap sphere. As dual screening tests can only be proceeded once $n \leq m$, it is possible that they can not be proceeded at the beginning of the algorithm. In such cases, they are triggered once the dimension has been enough reduced by primal screening tests.

Algorithm 1: Turbo screening

```
Input: x^0, A, y, λ
I_p^⋆ ← ∅, I_d^⋆ ← ∅
while true do
    x, Ax, A^T u ← MethodIteration(x, A, y, λ)
    I_p^⋆ ← I_p^⋆ ∪ PrimalScreening(Ax, A^T u, λ)
    x, A, A^T u ← ReduceDimension(I_p^⋆)
    if n ≤ m then
        α ← FindRelaxedValue()
        I_d^⋆ ← I_d^⋆ ∪ DualScreening(Ax, A^T u, α)
        (Ã^T Ã)^{-1} ← RankOneUpdate(I_d^⋆)
        x̃^⋆ ← SolveReducedProblem((Ã^T Ã)^{-1})
        if reduced problem optimum criterion is met then
            x^⋆ ← ReconstructSolution(x̃^⋆)
            return x^⋆
        end
    end
end
```

3.2 Implementation details

Some part of the algorithm must be done carefully in order to get the most out of Turbo screening.

Initial point and continuation : When one need to solve ($\mathcal{P}$) for a given value of λ, many ways allow to find a good starting point. However, it is hard to find one which really improve the convergence speed. Thus, the point $\mathbf{x}_0 = \mathbf{0}$ is often used in our algorithms. Nevertheless, in many applications, one need to solve ($\mathcal{P}$) for several regularization values, often placed in a logarithmic scale. If we denote $\mathbf{x}^\star_{\lambda^k}$ the solution of ($\mathcal{P}$) for the k-th regularization value, it is highly recommended to use this point as a starting point for the problem with the $(k+1)$-th regularization value. This continuation method is very efficient as if $\lambda \simeq \lambda'$, then $\mathbf{x}^\star_\lambda \simeq \mathbf{x}^\star_{\lambda'}$. Thus the initial point used for every regularization is a good starting point if the sequence of regularization is closely spaced.

Low computation cost for screening test : This point is crucial when using Turbo screening. To efficiently compute the duality gap, one only need the values of $\mathbf{A}\mathbf{x}$ and $\mathbf{A}^{\mathrm{T}}\mathbf{u}$. The first step is to compute the dual-scaled variable $\mathbf{u}_{\mathrm{f}}$ using (1.2.1b). When doing this operation, one can extract the scaling coefficient $s_{\mathbf{u}} := \max_i \mathbf{a}_i^{\mathrm{T}}\mathbf{u}$, which can easily be computed using the known value of $\mathbf{A}^{\mathrm{T}}\mathbf{u}$. The value $\mathbf{A}\mathbf{x}_{\mathrm{f}}$ costs nothing to compute. At this point, the values $\mathrm{P}(\mathbf{x}_{\mathrm{f}})$, $\mathrm{D}(\mathbf{u}_{\mathrm{f}})$ and $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$ can be easily computed using $\mathbf{A}\mathbf{x}_{\mathrm{f}}$. Then, $s_{\mathbf{u}}$ can be used to compute cheaply $\mathbf{A}^{\mathrm{T}}\mathbf{u}_{\mathrm{f}} = s_{\mathbf{u}}\mathbf{A}^{\mathrm{T}}\mathbf{u}$. Using $\mathbf{A}^{\mathrm{T}}\mathbf{u}_{\mathrm{f}}$ and $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}})$, the primal and dual gap screening tests can be proceeded at virtually no cost.

Handling dimension modification : When using primal screening tests, the dimension can be modified at every iteration. Every time that new components of $\mathcal{I}^\star_{\mathrm{p}}$ are screened, one must delete the corresponding coefficients in $\mathbf{x}$, $\mathbf{A}^{\mathrm{T}}\mathbf{u}$ and the corresponding column of $\mathbf{A}$. If dual screening tests are used, one also must update the value of $(\tilde{\mathbf{A}}^{\mathrm{T}}\tilde{\mathbf{A}})^{-1}$ if a unsafely dual-screened coefficient is finally identified as belonging to $\mathcal{I}^\star_{\mathrm{p}}$. This can be done with rank-one update rules [Kha08]. It is highly recommended to keep a track of the initial index of $\mathbf{x}_0$ in order to be able to reconstruct $\mathbf{x}^\star$ using (1.3.2).

Determine if $\mathcal{I}^\star_{\mathrm{d}}$ is well identified : One need to check if $\mathcal{I}^\star_{\mathrm{d}}$ is well identified by the heuristic dual screening test every time that the problem (1.3.1) is solved. One can choose to recompute the duality gap for the reduced problem using the optimal variable $\tilde{\mathbf{x}}^\star$. If $\mathcal{I}^\star_{\mathrm{d}}$ is indeed well identified, the duality gap is 0 and $\mathbf{x}^\star$ can be reconstructed using (1.3.2). Otherwise, one need to continue to proceed primal screening tests to correct dual screening mistakes. As $\mathrm{card}(\mathcal{I}^\star_{\mathrm{d}})$ is usually low, the cost of checking optimality of the reduced problem is negligible.

4 Numerical results

In this final section, we present some numerical results obtained with a Turbo screening approach. This approach is tested for FISTA, ADMM and the Frank-Wolfe (FW) algorithm. These algorithms are implemented as explained in appendix B and according 3 in order to use Turbo screening efficiently.

4.1 Experimental setup

In the following, the target accuracy on the dual gap is 10^{-15}. For a fixed value of n, various dictionary generation methods are tested. For each method, various values of m are tested. The continuation method mentioned in 3 for a range of regularization.

Methods The following methods are tested (*c.f.* appendix B for more details) :

- FISTA : A backtracking method with $\mu = 1.1$ is used.
- ADMM : The $\mathbf{x}$-update (B.2.11) is solved using a conjugate gradient method. Values for ρ-update in (B.2.7) are set to $\tau^{incr} = 1.1$, $\tau^{decr} = 1.1$, $\mu = 100$.
- FW algorithm : The improved version is used. Subproblems are solved using FISTA (with Turbo screening if enabled).

Dictionaries Five type of dictionary are used :

- Gaussian : Columns are sampled according to a Gaussian law. Low correlation between columns.
- DCT : Discrete Cosine Transom (type 1) matrix of shape $\max(m, n) \times \max(m, n)$. Lines are swapped randomly and the matrix is truncated to fit a dimension $m \times n$.
- Toeplitz : Toeplitz matrix of shape $\max(m, n) \times \max(m, n)$ with the constant column sampled randomly. Matrix is truncated to fit a dimension $m \times n$.
- Uniform : Columns are sampled according to an Uniform law. High correlation between columns.
- Constant-noise : Constant matrix with coefficients equal to 2 plus a Gaussian noise. High correlation between columns.

The observation vector $\mathbf{y}$ is always sampled according to a Gaussian law.

Screening tests Primal screening tests are proceeded with the primal gap sphere (2.1.3). Unsafe dual screening tests are proceeded with the dual gap sphere (2.2.3). In FISTA, α is the backtracked Lipschitz constant. In ADMM and in FW, $\alpha = \min(\|\mathbf{A}\|_1, \|\mathbf{A}\|_\infty)$

Comparison criterion For each method, we record the number of operations needed to reach the targeted dual gap. Only the number of multiplications is taken into account as additions have a marginal cost compared to multiplications in practice. A gain equal to 2 means that Turbo screening needs twice less operations than the method with primal screening only.

Repeats Experiments were made using INRIA's cluster *IGRIDA*. For each different configuration, 50 repetitions were made in order to reduce the variance in the results.

4.2 Convergence scheme in extreme cases

Using the numerical results which can be found in appendix A, we extract for each algorithm one good and one bad setup to use Turbo screening. The notation `+pst` and `+dst` means respectively that primal screening tests and dual screening tests are used. Thus, when the method name is appended with `+pst+dst`, Turbo screening is enabled.

4.2.1 FISTA

For dictionaries with low correlation between columns, Turbo screening seems to have a higher gain when the dictionary is *tall* (*i.e.* $n >> m$). As $\text{card}(\mathcal{I}_{\text{d}}^{\star}) \leq m$, identifying $\mathcal{I}_{\text{d}}^{\star}$ is even faster as this set has few elements. Thus, the greedy active set heuristic identifies quickly the right set and the subproblem resolution allows to reach a high precision and stop the algorithm very soon, leading to a higher gain. When there is more correlation between columns, Turbo screening seems to improve the method for any dictionaries. However, the gain the higher for *flat* dictionaries (*i.e.* $m >> n$). When a dictionary is flat, dual screening tests are enabled from the top of the algorithm, making Turbo screening pretty aggressive and fast to identify $\mathcal{I}_{\text{d}}^{\star}$, whereas for tall dictionaries, Turbo screening is only enabled when the dimension is enough reduced by primal screening tests. One can notice that against a raw FISTA (without screening at all), the gain using Turbo screening is really significant and goes up to 150 times the number of operations.

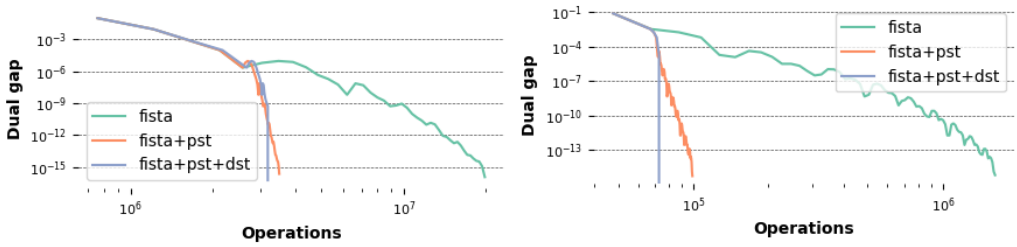

Figure 10: Left : Bad setup for Turbo screening with FISTA (500×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.6$). Right : Good setup for Turbo screening with FISTA (20×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.6$).

4.2.2 ADMM

In ADMM, the behaviour is really different between tall and flat dictionaries. When $\mathbf{A}$ is tall, primal screening tests allow ADMM to converge really fast once the dimension is enough reduced (*i.e.* $n \leq m$) by primal screening tests. Indeed, once this state is met, the problem becomes strongly-convex, making ADMM to converge very fast because it has already removes null coefficient of the solution. However, when $\mathbf{A}$ is flat, Turbo screening is enabled from the top of the algorithm and seems particularly aggressive when identifying $\mathcal{I}_{\mathrm{d}}^{\star}$. In the right graphic, only 3 iterations are needed to reach the target precision with Turbo screening. The gain goes up to 10 times the number of operations against a raw ADMM.

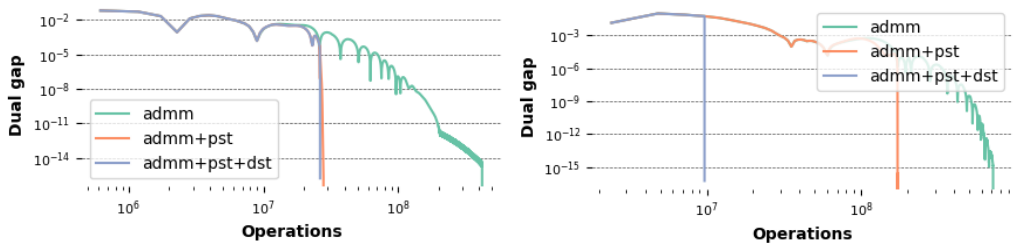

Figure 11: Left : Bad setup for Turbo screening with ADMM (100×300 uniform dictionary with $\lambda/\lambda_{\max} = 0.6$). Right : Good setup for Turbo screening with ADMM (500×300 uniform dictionary with $\lambda/\lambda_{\max} = 0.6$).

4.2.3 Frank-Wolfe

The improvement used in FW and explained in appendix also allows to determine the right support of $\mathbf{x}^{\star}$ during the algorithm. The dual gap first decreases slowly and once the support is well identified, the subproblem resolution allows to extract the right solution with a very high accuracy. Because of the slow decreasing of the dual gap, screening tests are not very efficient. That is why there is no significant improvement when primal screening tests are used. However, is seem that dual screening tests allow to identify $\mathcal{I}_{\mathrm{d}}^{\star}$ faster than FW, thus it improved the convergence. When $\mathbf{A}$ is tall, $\mathcal{I}_{\mathrm{d}}^{\star}$ is small so the gain brought by Turbo screening is even more significant as the identification of this set is faster. The gain goes up to 150 times the number of operations, even if FW is used with primal screening tests.

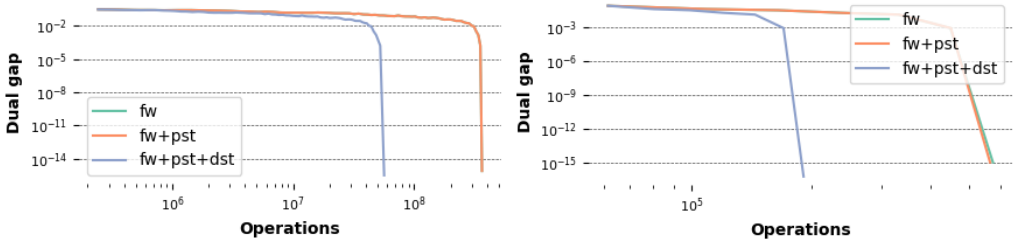

Figure 12: Left : Bad setup for Turbo screening with FW (200×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.2$). Right : Good setup for Turbo screening with FW (50×300 Gaussian dictionary with $\lambda/\lambda_{\max} = 0.6$).

4.3 Heuristic dual screening performance

Finally, we wanted to estimate the safeness of the relaxed sphere used for dual screening. As this sphere is unsafe, it can produce mistakes when identifying $\mathcal{I}_{\mathrm{d}}^{\star}$ (which are corrected by safe primal screening during the algorithm). For each algorithm we record the percentage of wrongly dual screened coefficient (*wdsc* in the graphic) over the n coefficients. One can notice that there are more mistakes when the dictionary is flat because card$(\mathcal{I}_{\mathrm{d}}^{\star})$ is larger. Thus, more mistakes can occur and primal screening tests do not allow to remove all the tricky coefficients as Turbo screening is enabled from the top of the algorithm. We can notice that very few mistakes are made for dictionaries with high correlation between columns. With such dictionaries, convergence is slow at the beginning of the algorithm. Thus, dual screening has a bad screening rate in the first iterations. Then, the gap is tightened, and dual screening become to identify $\mathcal{I}_{\mathrm{d}}^{\star}$ but tricky coefficients have already been identified by primal screening tests so the risk of screening mistakes is minimized. In general, no more than 2% of the coefficients are wrongly dual screened.

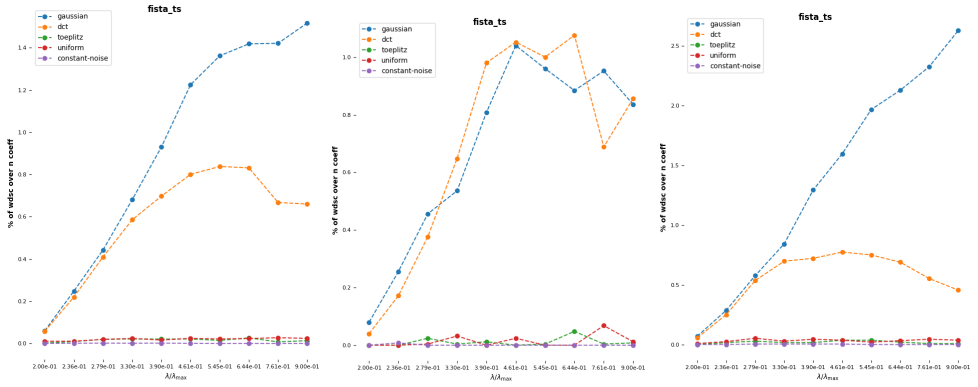

Figure 13: Percentage of wrongly dual screened coefficient over n coefficients in FISTA. Left : mean over all the dimension tested. Center : Flat dictionary 100×300. Right : Tall dictionary 500×300.

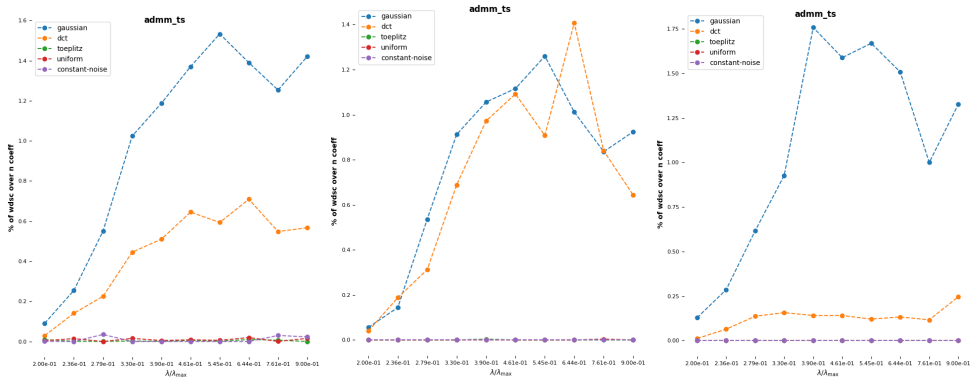

Figure 14: Percentage of wrongly dual screened coefficient over n coefficients in ADMM. Left : mean over all the dimension tested. Center : Flat dictionary 100×300. Right : Tall dictionary 500×300.

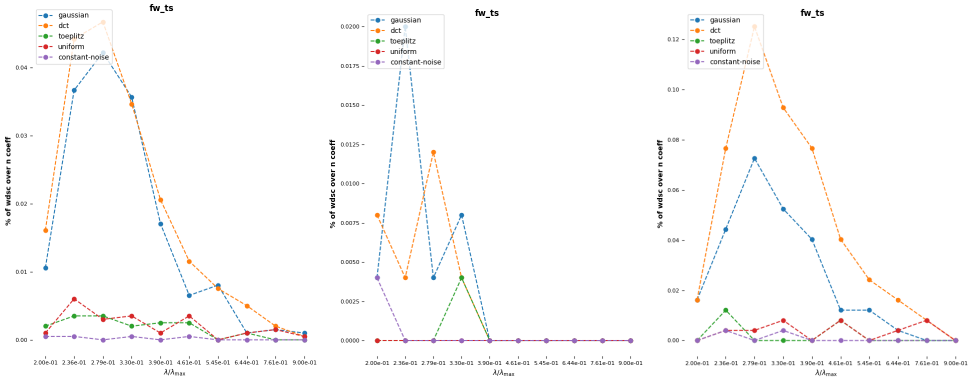

Figure 15: Percentage of wrongly dual screened coefficient over n coefficients in FW. Left : mean over all the dimension tested. Center : Flat dictionary 100×300. Right : Tall dictionary 500×300.

5 Conclusion

For the LASSO, some setups can give a hard time for most of solving methods. Primal screening tests, proceeded in the dual space, already existed to face dimensionality issues. We created new dual screening tests, proceeded in the primal space, in order to bypass the last iterations of most of optimization methods. As these last iterations are usually not very effective in getting close to the solution, reaching a high precision is usually very costly and many iterations are needed to improve a bit the current precision. Our new screening tests allow to sidestep this issue by solving an auxiliary linear problem of a low dimension, used to reconstruct the optimal solution. As linear problems can be solved with a machine-epsilon accuracy, Turbo screening effectively addresses the problem when the problem needs to be solved accurately. Even if dual screening test are hard to proceed in practice because some values needed can be costly to compute, heuristics can be used. There is still some work to do in finding new dual screening regions, cheaper to construct or with a better screening rate. It could be very interesting to generalize primal and dual screening tests to more general convex and constrained problems. In this way, it could be possible to create a general solver proceeding a primal-dual identification of the active sets and allowing to simply solve an unconstrained problem once these sets have been identified. Transforming a contained problem into an unconstrained problem allows to use far more efficient methods to recover the solution.

A Numerical results

A.1 FISTA

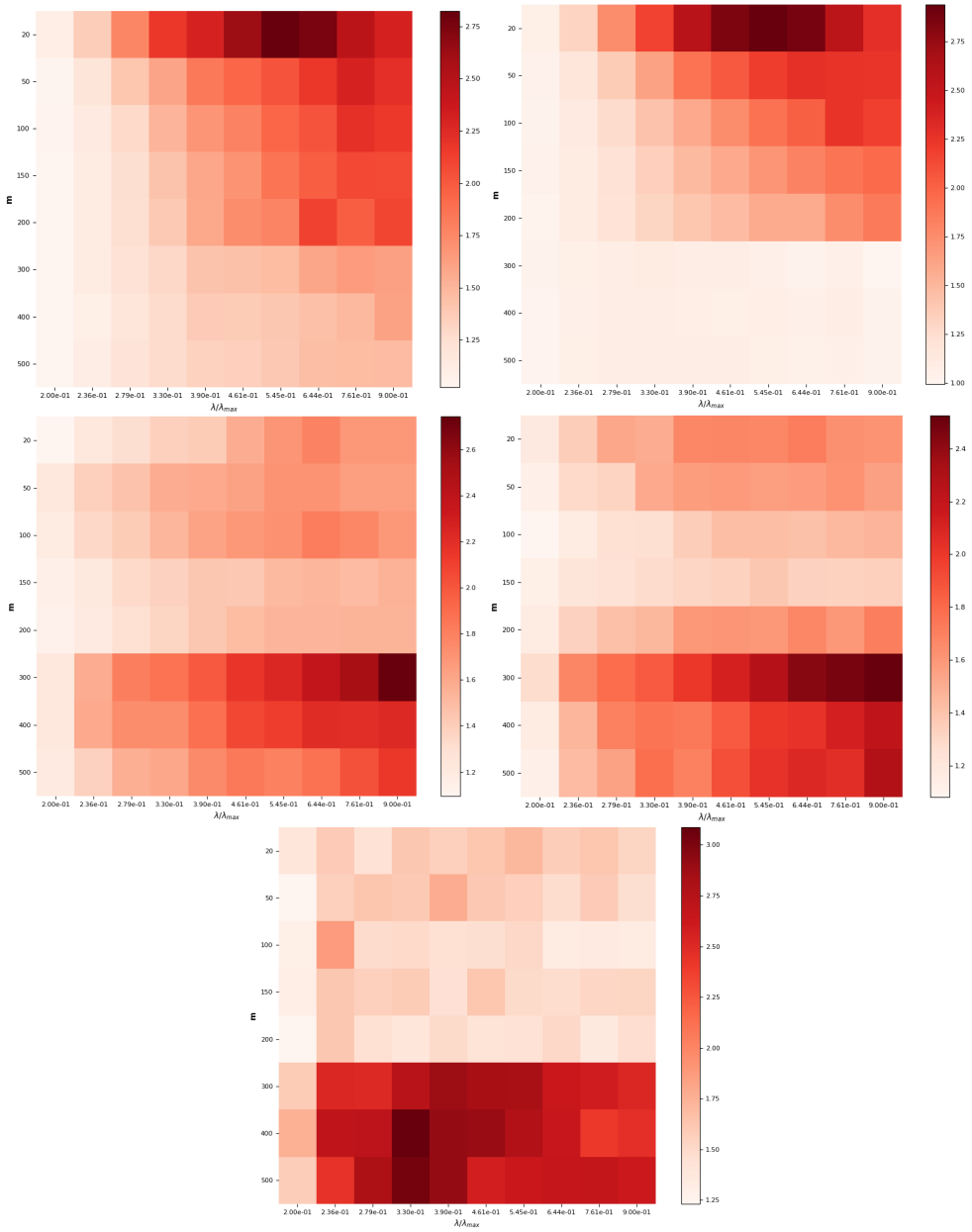

Figure 16: Gain with FISTA for Gaussian, DCT, Toeplitz, Uniform and Constant-noise dictionaries (from top left to bottom right) with $n = 300$. Gain is capped to 5.

A.2 ADMM

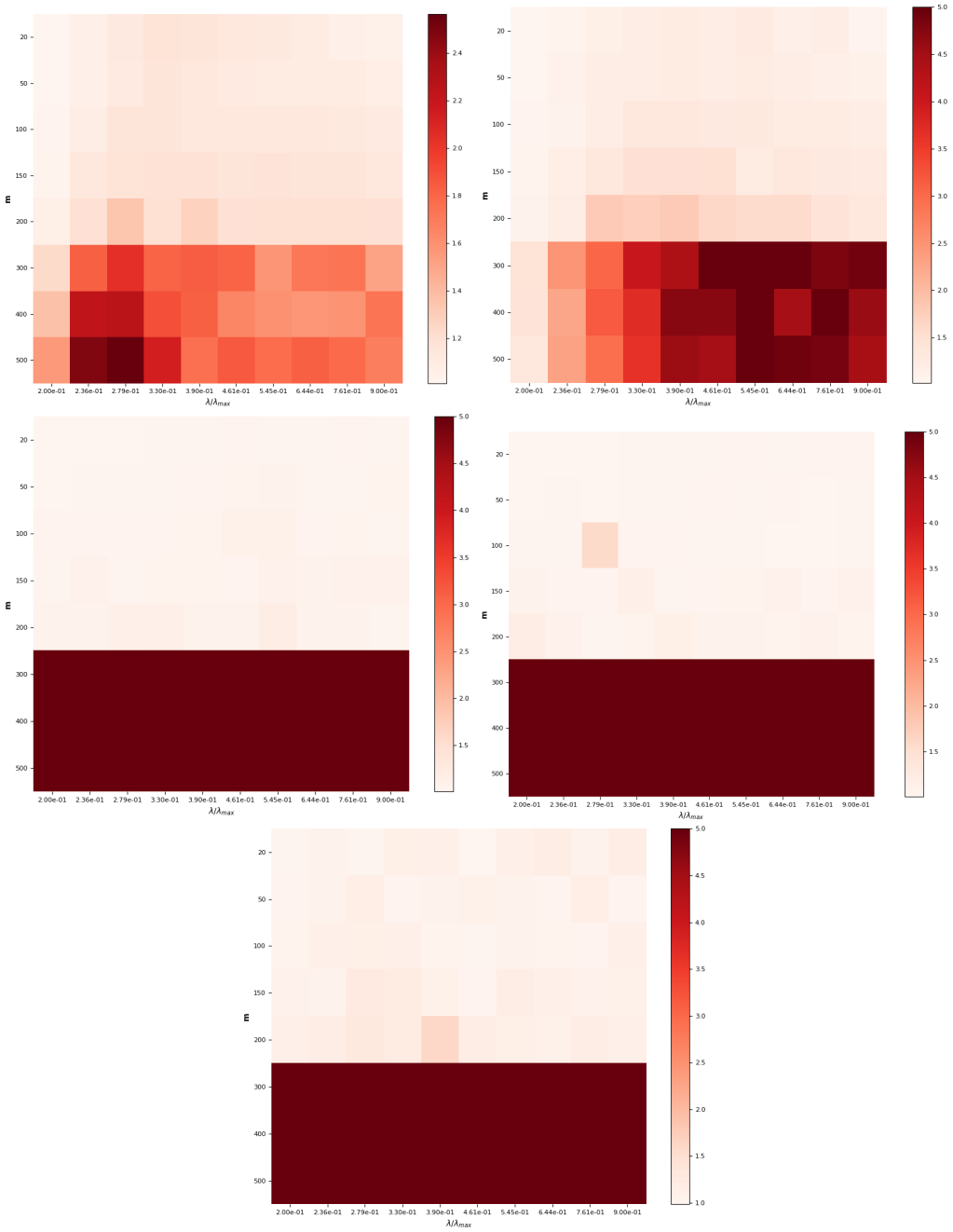

Figure 17: Gain with ADMM for Gaussian, DCT, Toeplitz, Uniform and Constant-noise dictionaries (from top left to bottom right) with $n = 300$. Gain is capped to 5.

A.3 FW

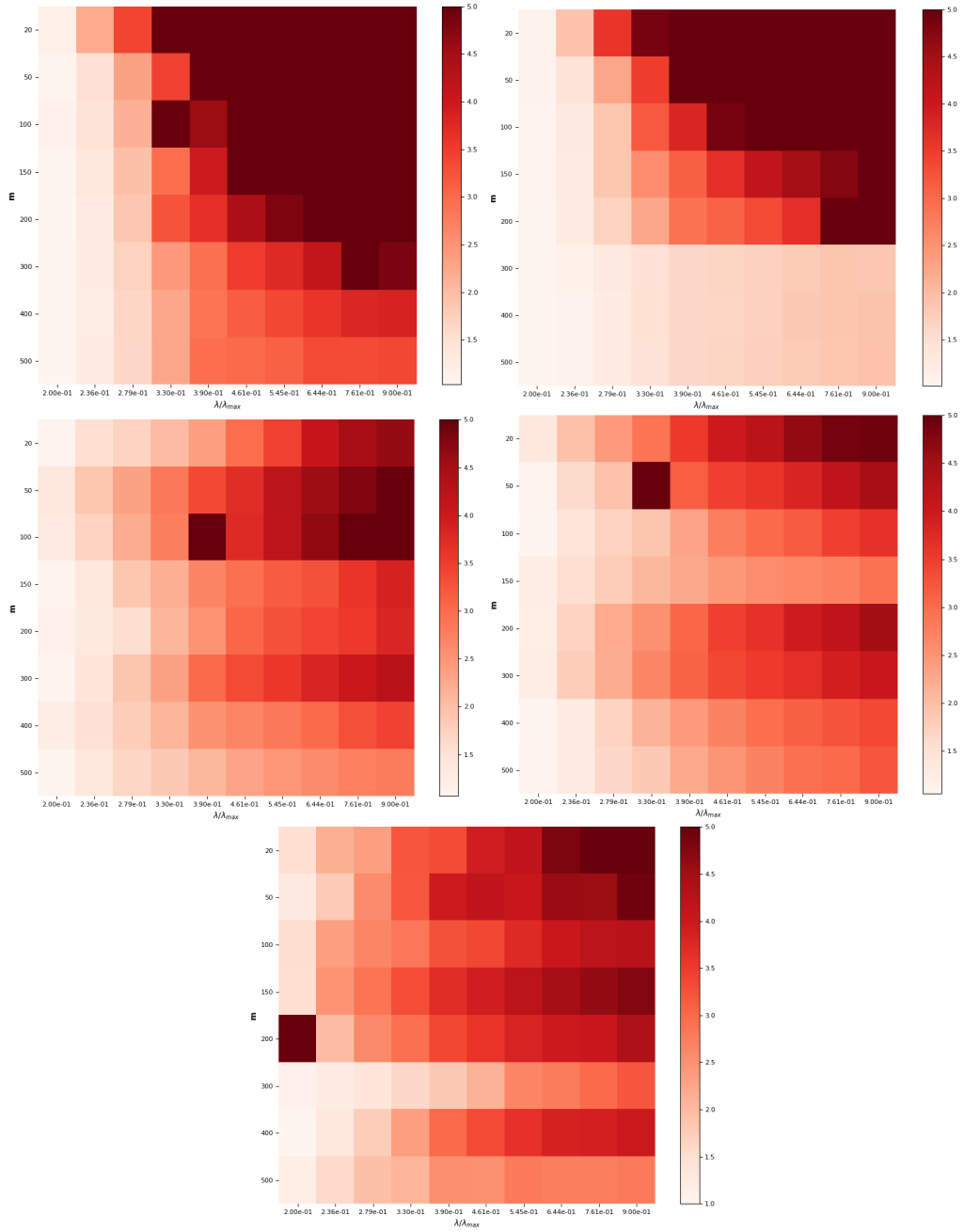

Figure 18: Gain with FW for Gaussian, DCT, Toeplitz, Uniform and Constant-noise dictionaries (from top left to bottom right) with $n = 300$. Gain is capped to 5.

B Optimization methods

Many efficient methods allow to tackle ($\mathcal{P}$). As this problem has grown in importance over the past decades, these methods have been improved in many ways. In the following, we present the basic principles of some of these.

B.1 FISTA

FISTA [BT09] is probably the most famous algorithm allowing to solve the LASSO. As it is a gradient-based method, it is particularly well suited for the Turbo screening approach. FISTA is mostly used to solve the classical LASSO problem, and in the following, we derive how to use it in the case of the non-negative LASSO problem. We explain how to recover the value of $\nabla \mathrm{P}$ during the iterations to efficiently proceed screening tests.

B.1.1 Basic step

The FISTA algorithm is suited for problems aiming to minimize $\mathrm{H}(\mathbf{x}) = \mathrm{P}(\mathbf{x}) + \mathrm{G}(\mathbf{x})$ where P is a smooth and convex function and G is a convex function, possibly non-smooth. We can re-write ($\mathcal{P}$) as

$$\mathbf{x}^\star \in \underset{\mathbf{x}}{\operatorname{argmin}} \Big\{ \mathrm{H}(\mathbf{x}) = \underbrace{\tfrac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 + \lambda \mathbf{1}^{\mathrm{T}}\mathbf{x}}_{\mathrm{P}(\mathbf{x})} + \underbrace{\mathbb{I}_{\{\mathbf{x} \geq \mathbf{0}\}}(\mathbf{x})}_{\mathrm{G}(\mathbf{x})} \Big\} \tag{B.1.1}$$

where $\mathbb{I}_{\mathcal{X}}$ is the convex characteristic function of the set $\mathcal{X}$. FISTA consists in the following steps, which are proceeded until the stopping criterion is met :

$$\mathbf{x}_k = p_L(\mathbf{z}_k) \tag{B.1.2a}$$

$$t_{k+1} = \left(1 + \sqrt{1 + 4t_k^2}\right)/2 \tag{B.1.2b}$$

$$\mathbf{z}_{k+1} = \mathbf{x}_k + \tfrac{t_k - 1}{t_{k+1}}(\mathbf{x}_k - \mathbf{x}_{k-1}) \tag{B.1.2c}$$

with $\mathbf{z}_0 = \mathbf{x}_0$, $t_0 = 1$ and where L is a Lipschitz constant of $\nabla \mathrm{P}$. The operator $p_{.}(.)$ is defined as

$$\begin{aligned} p_L(\mathbf{z}) &= \underset{\mathbf{x}}{\operatorname{argmin}} \left\{ \mathrm{G}(\mathbf{x}) + \tfrac{L}{2} \left\| \mathbf{x} - \left(\mathbf{z} - \tfrac{1}{L}\nabla \mathrm{P}(\mathbf{z}) \right) \right\|_2^2 \right\} \\ &= \max \left\{ \mathbf{0} \,,\, \mathbf{z} - \tfrac{1}{L}\nabla \mathrm{P}(\mathbf{z}) \right\} \end{aligned} \tag{B.1.3}$$

FISTA has a quadratic rate of convergence and both sequences $\{\mathbf{x}_k\}_k$ and $\{\mathbf{z}_k\}_k$ converge toward $\mathbf{x}^\star$.

B.1.2 Backtracking

L must be a Lipschitz constant of $\nabla \mathrm{P}$ so one has $L \geq \lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ for (B.1.1). However, computing eigenvalues of $\mathbf{A}^{\mathrm{T}}\mathbf{A} \in \mathbb{R}^{n \times n}$ can be very costly. We can

compute a suitable value for L with the following backtracking rule. Let i_k the smallest integer at the iteration k such that with $\bar{L} = \mu^{i_k} L_{k-1}$, we have

$$\mathrm{P}(p_{\bar{L}}(\mathbf{x}_{k-1})) \leq \mathrm{Q}_{\bar{L}}(p_{\bar{L}}(\mathbf{x}_{k-1}), \mathbf{x}_{k-1}) \tag{B.1.4}$$

for a given $\mu > 1$ and where $\mathrm{Q}_L(\mathbf{x}, \mathbf{z}) = \mathrm{P}(\mathbf{z}) + \langle \mathbf{x} - \mathbf{z}, \nabla f(\mathbf{z}) \rangle + L/2 \, \|\mathbf{x} - \mathbf{z}\|_2^2 + \mathrm{G}(\mathbf{x})$. μ is called the *backtracking parameter*. Then, one can set $L = \mu^{i_k} L_{k-1}$. We can start with any $L_0 > 0$ but as we work with dictionary with normalized columns, one has $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A}) \geq 1$, so a suitable starting value is $L = 1$.

During the algorithm, the dimension may be reduced with primal screening tests. Using *Cauchy's interlacing theorem* [Hwa04], we know that the value of $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$ is non-increasing. Thus, we can either re-initialize the backtracked L to L_0 every time the dimension is updated and continue to use backtracking if needed or keep its actual value for the rest of the algorithm.

B.1.3 Taking advantage of FISTA

The key advantage of FISTA is that it has the good properties mentioned in subsection 1.4. The value of $\nabla \mathrm{P}(\mathbf{z}_k)$ can be used to compute the duality gap and to proceed screening tests with a low additional computational cost. Furthermore, the duality gap is often used as stopping criterion so it is computed most of the time anyway. We can also notice that the duality gap tends to 0 quickly in FISTA. The screening tests will be even more strengthened during the iterations.

B.2 ADMM

Alternating Direction Method of Multipliers (ADMM) [BPC11], also known as Douglas-Rachford slitting, is a method allowing to solve problems where constraints split. It is also often used to solve ($\mathcal{P}$).

B.2.1 Problem and basic step

We consider problems of the following form :

$$\min_{\mathbf{x},\mathbf{z}} \{\mathrm{P}(\mathbf{x}) + \mathrm{G}(\mathbf{z})\} \qquad \text{s.t.} \quad \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} = \mathbf{c} \tag{B.2.1}$$

where P and G are convex functions. The only difference from a general linear equality-constrained convex problem is that the initial variable has been split into two parts, namely $\mathbf{x}$ and $\mathbf{z}$. For (B.2.1), the *augmented Lagrangian* is

$$\mathcal{L}_\rho(\mathbf{x}, \mathbf{z}, \mathbf{t}) = \mathrm{P}(\mathbf{x}) + \mathrm{G}(\mathbf{z}) + \mathbf{t}^{\mathrm{T}}(\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} - \mathbf{c}) + \tfrac{\rho}{2} \|\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z} - c\|_2^2 \tag{B.2.2}$$

with $\rho > 0$. ADMM consists in the iterations

$$\mathbf{x}_{k+1} = \operatorname*{argmin}_{\mathbf{x}} \mathcal{L}_\rho(\mathbf{x}, \mathbf{z}_k, \mathbf{t}_k) \tag{B.2.3a}$$

$$\mathbf{z}_{k+1} = \operatorname*{argmin}_{\mathbf{z}} \mathcal{L}_\rho(\mathbf{x}_{k+1}, \mathbf{z}, \mathbf{t}_k) \tag{B.2.3b}$$

$$\mathbf{t}_{k+1} = \mathbf{t}_k + \rho(\mathbf{A}\mathbf{x}_{k+1} + \mathbf{B}\mathbf{z}_{k+1} - \mathbf{c}) \tag{B.2.3c}$$

The vector $\mathbf{t}$ acts as the dual variable of (B.2.1). The scaled version of ADMM is often used as it contains sorter update formulas. This version is obtain by reorganizing (B.2.2) :

$$\mathbf{x}_{k+1} = \underset{\mathbf{x}}{\operatorname{argmin}}\ \mathrm{P}(\mathbf{x}) + \tfrac{\rho}{2}\left\|\mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{z}_k - \mathbf{c} + \mathbf{t}_k\right\|_2^2 \tag{B.2.4a}$$

$$\mathbf{z}_{k+1} = \underset{\mathbf{z}}{\operatorname{argmin}}\ \mathrm{G}(\mathbf{x}) + \tfrac{\rho}{2}\left\|\mathbf{A}\mathbf{x}_{k+1} + \mathbf{B}\mathbf{z} - \mathbf{c} + \mathbf{t}_k\right\|_2^2 \tag{B.2.4b}$$

$$\mathbf{t}_{k+1} = \mathbf{t}_k + \mathbf{A}\mathbf{x}_{k+1} + \mathbf{B}\mathbf{z}_{k+1} - \mathbf{c} \tag{B.2.4c}$$

In practice, ADMM converges to modest accuracy in few iterations and this is sufficient for many applications. However, it can struggle to reach a high accuracy.

Stopping criterion For (B.2.1), optimality conditions are

$$\mathbf{A}\mathbf{x}^\star + \mathbf{B}\mathbf{z}^\star - \mathbf{c} = 0 \tag{B.2.5a}$$

$$0 \in \partial f(\mathbf{x}^\star) + \mathbf{A}^{\mathrm{T}}\mathbf{t}^\star \tag{B.2.5b}$$

$$0 \in \partial g(\mathbf{z}^\star) + \mathbf{B}^{\mathrm{T}}\mathbf{t}^\star \tag{B.2.5c}$$

Were $\mathbf{t}^\star$ is the dual optimal variable. The following quantities can be viewed as the primal and the dual residuals for (B.2.5b) and (B.2.5c) :

$$r_{k+1} = \mathbf{A}\mathbf{x}^\star + \mathbf{B}\mathbf{z}^\star - \mathbf{c} \tag{B.2.6a}$$

$$s_{k+1} = \rho\mathbf{A}^{\mathrm{T}}\mathbf{B}(\mathbf{z}_{k+1} - \mathbf{z}_k) \tag{B.2.6b}$$

Thus, we can set a primal and a dual precision ϵ^{pri} and ϵ^{du} on $\|r_{k+1}\|$ and $\|s_{k+1}\|$ to stop the method. In [BPC11], a way to set a relative and an absolute precision on these values is also given.

Variations Two main improvements can be added to the classical ADMM algorithm. The first one consists in tuning ρ during the iterations using three parameters τ^{incr}, τ^{decr} and μ as following :

$$\rho^{k+1} = \begin{cases} \rho_k\tau^{incr} & \text{if } \|r_k\|_2 > \mu\,\|s_k\|_2 \\ \rho_k/\tau^{decr} & \text{if } \|s_k\|_2 > \mu\,\|r_k\|_2 \\ \tau_k & \text{otherwise} \end{cases} \tag{B.2.7}$$

The idea is to keep the primal and dual residuals close to each other and make them converging to 0 at the same speed. The second improvement is the introduction of a relaxation parameter $\alpha \in]0, 2[$ during the $\mathbf{z}$ and $\mathbf{t}$ updates. The quantity $\mathbf{A}\mathbf{x}_{k+1}$ can be replaced by

$$\alpha\mathbf{A}\mathbf{x}_{k+1} - (1-\alpha)(\mathbf{B}\mathbf{z}_k - \mathbf{c}) \tag{B.2.8}$$

to improve the convergence speed.

B.2.2 Application to the non-negative LASSO

($\mathcal{P}$) can also be parametrized as

$$\mathbf{x}^\star \in \underset{\mathbf{x},\mathbf{z}}{\operatorname{argmin}} \Big\{ \underbrace{\tfrac{1}{2}\,\|\mathbf{y}-\mathbf{A}\mathbf{x}\|_2^2 + \lambda \mathbf{1}^{\mathrm{T}}\mathbf{x}}_{\mathrm{P}(\mathbf{x})} + \underbrace{\mathbb{I}_{\{\mathbf{z}\geq \mathbf{0}\}}(\mathbf{z})}_{\mathrm{G}(\mathbf{z})} \Big\} \qquad \text{s.t.} \qquad \mathbf{x}=\mathbf{z} \tag{B.2.9}$$

in order to be solved with ADMM. As G has a particular expression, the $\mathbf{z}$-update (B.2.4b) is simple to proceed :

$$\mathbf{z}_{k+1} = \max(0, \mathbf{x}_{k+1} + \mathbf{t}_k) \tag{B.2.10}$$

However, the solution of the $\mathbf{x}$-update (B.2.4a) is

$$\mathbf{x}_{k+1} = (\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}(\tfrac{1}{\rho}(\mathbf{z}_k - \mathbf{t}_k) + \mathbf{A}^{\mathrm{T}}\mathbf{y} - \lambda\mathbf{1}) \tag{B.2.11}$$

Usually, the value of $\mathbf{A}^{\mathrm{T}}\mathbf{y}$ is computed at the beginning of the algorithm and a $\mathbf{LU}$ factorization of $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ can be computed to diminish the computational cost. However, when primal screening tests are used, we have to handle the change of dimension. The value of $\mathbf{A}^{\mathrm{T}}\mathbf{y}$ can still be pre-computed and reduced when the dimension is modified. However, we need to recompute a $\mathbf{LU}$ factorization at each modification of dimension. Thus it is preferable to compute directly $(\mathbf{A}^{\mathrm{T}}\mathbf{A})^{-1}$ and to used rank-one rules to update this inverse [Kha08] every time the dimension is modified or to use a conjugate gradient method [BHM98] to solve (B.2.11).

B.2.3 Taking advantage of ADMM

ADMM also satisfy conditions given in paragraph 1.4. Using a conjugate gradient method for the $\mathbf{x}$-update, we can get the value of $\nabla\mathrm{P}$ needed for screening tests with no additional computational cost. As in FISTA, we have $\mathrm{gap}(\mathbf{x}_{\mathrm{f}}, \mathbf{u}_{\mathrm{f}}) \to 0$ relatively quickly so screening tests are strengthened during the iterations and below a certain dual gap, $\mathcal{I}_{\mathrm{p}}^\star$ and $\mathcal{I}_{\mathrm{d}}^\star$ can be fully identified.

B.3 Frank-Wolfe

Frank-Wolfe (FW) algorithm [Jag13], also known as the Conditional gradient algorithm, is one of the oldest algorithm developed for convex optimization. Although it is known to have bad performances, recent improvements allows it to be competitive with other methods.

B.3.1 Basic step

The FW algorithm is suited for problems expressed as

$$\min_{\mathbf{x}\in\mathcal{C}} \mathrm{P}(\mathbf{x}) \tag{B.3.1}$$

where $\mathcal{C}$ is compact and convex and P is convex and differentiable. The principle is simple : linearise P near the current point, find a descend direction $\mathbf{d}$ and a

step size γ by line each to compute a new point. FW consists in the following iterations :

$$\mathbf{d} = \underset{\mathbf{s}\in\mathcal{C}}{\operatorname{argmin}} \langle \nabla \mathrm{P}(\mathbf{x}_k), \mathbf{d} \rangle \tag{B.3.2a}$$

$$\mathbf{x}_{k+1} = \gamma \mathbf{x}_k + (1-\gamma)\mathbf{d} \tag{B.3.2b}$$

where $\gamma = \arg\min_{\tilde{\gamma}\in[0,1]} \mathrm{P}(\tilde{\gamma}\mathbf{x}_k + (1-\tilde{\gamma})\mathbf{d})$. We can notice that if $\mathcal{C}$ is not compact, the line search may tends to infinity. If $\nabla\mathrm{P}$ is Lipschitz, FW usually converges in sub-linear time [NW06].

B.3.2 Improvements

In practice, the convergence of FW can be very slow for problems such as ($\mathcal{P}$). Indeed, the step (B.3.2b) is a convex combination so if the algorithm moves toward a direction of a null coefficient of $\mathbf{x}^\star$, it is really difficult to correct this mistake in the following iterations. For this reason, a significant improvement was introduced to tackle problems such as ($\mathcal{P}$) [Jag13]. The principle is to assume that the algorithm will moves most of the time toward directions of non-null coefficients of $\mathbf{x}^\star$. Thus, it is possible to estimate the support of the solution during the iterations and to solve subproblems such as (1.3.1) but still with with positivity constraints. If the support is well identified, the subproblem allow to recover the optimal solution. In this way, FW acts as an Active-set method [NW06]. This allows to converge much faster than without the improvement if $\mathrm{supp}(\mathbf{x}^\star)$ is small. If this improvement is combined with primal screening tests, allowing to identify elements outside the support, this diminish the risk of mistakes in the support identification.

B.3.3 Application to non-negative LASSO

In ($\mathcal{P}$), the feasible region is not bounded. However, one can show that $\|\mathbf{x}^\star\|_2 \leq \|\mathbf{y}\|_2^2 / 2\lambda$ so the feasible region can be reduced to the positive orthant of the sphere $\mathcal{S}(\mathbf{0}, \|\mathbf{y}\|_2^2/2\lambda)$, which is a convex and compact region. We can find the descend direction in (B.3.2a) as a vector collinear to the i-th canonical basis vector where $i = \arg\min_j [\mathbf{A}^\mathrm{T}(\mathbf{A}\mathbf{x}-\mathbf{y})]_j$. The solution of (B.3.2b) also admits a closed form. Every time that a new component of the support is identified, the reduced subproblem in the improved method can be solved with any method. As $|\mathcal{I}_\mathrm{d}^\star|$ is usually small, subproblems are solved quickly. FW has the good properties mentioned in 1.4 as $\nabla\mathrm{P}$ can be extracted from (B.3.2a) and because $\mathrm{gap}(\mathbf{x}_\mathrm{f}, \mathbf{u}_\mathrm{f}) \to 0$ during the iterations.

B.4 Active-set method

Active-set (AS) method is suited for general quadratic programs and the particular structure of ($\mathcal{P}$) allows some simplifications in this method. It does not satisfy conditions given in 1.4 but it is also a method allowing to get $\mathbf{x}^\star$ with a machine epsilon precision. In the final Turbo screening method, we present and heuristic which is similar to AS.

B.4.1 General problem and theory

The AS method can be applied to convex quadratic problems with inequality constraints defined as

$$\min_{\mathbf{x}} \left\{ q(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^{\mathrm{T}}\mathbf{Q}\mathbf{x} + \mathbf{x}^{\mathrm{T}}\mathbf{c} \right\} \qquad \text{s.t.} \quad \mathbf{a}_i^{\mathrm{T}}\mathbf{x} \geq \mathbf{b}_i, \ \forall i \in \mathcal{I} \tag{B.4.1}$$

where $\mathbf{Q}$ is a symmetric semi-positive definite matrix. One needs $\{\mathbf{a}_i\}_{i\in\mathcal{I}}$ to be independent vectors. We note $\mathcal{A}(\mathbf{x}^\star) = \{i \in \mathcal{I}, \mathbf{a}_i^{\mathrm{T}}\mathbf{x}^\star = \mathbf{b}_i\}$ the saturated constraints at optimality. This set is called the *optimal active set* of (B.4.1). Minimizing q with the constraints $\mathbf{a}_i^{\mathrm{T}}\mathbf{x} = \mathbf{b}_i, \ \forall i \in \mathcal{A}(\mathbf{x}^\star)$ also leads to the optimal solution of (B.4.1).

AS method minimizes q with a subset $\mathcal{W}_{\mathrm{k}}$ (the *working set*) of the constraints imposed as saturated. At each iteration, it finds which constraints are missing or which are not necessary in $\mathcal{W}_{\mathrm{k}}$ to update this subset until $\mathcal{W}_{\mathrm{k}} = \mathcal{A}(\mathbf{x}^\star)$. The first working set $\mathcal{W}_0$ is a subset of the active constraints of the initial point $\mathbf{x}_0$.

We denote $\alpha_k\mathbf{p}_k = \mathbf{x}_{k+1} - \mathbf{x}_k$ the step between the new and the previous point and $\mathbf{g}_k = \nabla q(\mathbf{x}_k) = \mathbf{Q}\mathbf{x}_k + \mathbf{c}$. The subproblem solved at each iteration allowing to get the new point is

$$\mathbf{p}_k = \operatorname*{argmin}_{\mathbf{p}} \left\{ \tfrac{1}{2}\mathbf{p}^{\mathrm{T}}\mathbf{Q}\mathbf{p} + \mathbf{p}^{\mathrm{T}}\mathbf{c} \right\} \qquad \text{s.t.} \quad \mathbf{a}_i^{\mathrm{T}}\mathbf{p} = 0, \ \forall i \in \mathcal{W}_{\mathrm{k}} \tag{B.4.2}$$

We have $(\mathbf{p}_k)_i = (\mathbf{g}_k)_i$ and $(\mathbf{p}_k)_i = 0$ if $i \notin \mathcal{W}_{\mathrm{k}}$. Note that we have $\mathbf{a}_i^{\mathrm{T}}(\mathbf{x}_k + \alpha\mathbf{p}_k) = \mathbf{a}_i^{\mathrm{T}}\mathbf{x}_k = \mathbf{b}_i$ for all α if $i \in \mathcal{W}_{\mathrm{k}}$ if $\mathbf{x}_0$ is well chosen.

First suppose that $\mathbf{p}_k \neq \mathbf{0}$, we need to decide the step length $0 \leq \alpha_k \leq 1$. If $\mathbf{x}_k + \mathbf{p}_k$ is feasible, we can set $\alpha_k = 1$ and the new point is $\mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{p}_k$. Otherwise, we need α_k such that the new point is still feasible. We can set

$$\alpha_k = \min\left(1, \min_{i\notin\mathcal{W}_{\mathrm{k}}, \mathbf{a}_i^{\mathrm{T}}\mathbf{p}_k<0} \frac{\mathbf{b}_i - \mathbf{a}_i^{\mathrm{T}}\mathbf{x}_k}{\mathbf{a}_i^{\mathrm{T}}\mathbf{p}_k}\right) \tag{B.4.3}$$

This ensure that we set α_k to the largest value in $[0, 1]$ for which $\mathbf{x}_{k+1}$ is still feasible. The constraint i for which the minimum of (B.4.3) is reached is called the *blocking constraint*. The working set is updated as $\mathcal{W}_{k+1} = \mathcal{W}_{\mathrm{k}} \cup \{i\}$ (if $\alpha_k = 1$ then $\mathcal{W}_{k+1} = \mathcal{W}_{\mathrm{k}}$).

Now suppose that $\mathbf{p}_k = \mathbf{0}$. In this case, $\mathbf{x}_{k+1}$ minimizes q over the current working set. To check if this is also the solution of (B.4.1) (*i.e.* $\mathcal{W}_{\mathrm{k}} = \mathcal{A}(\mathbf{x}^\star)$), we can use the Lagrange multipliers $\hat{\lambda}$ of (B.4.1). Since, $\mathbf{p}_k = \mathbf{0}$, optimality conditions yield

$$\sum_{i\in\mathcal{W}_{\mathrm{k}}} \mathbf{a}_i\hat{\lambda}_i = \mathbf{g}_k^{\mathcal{W}_{\mathrm{k}}} = \mathbf{Q}^{\mathcal{W}_{\mathrm{k}}}\mathbf{x}_k^{\mathcal{W}_{\mathrm{k}}} + \mathbf{c}^{\mathcal{W}_{\mathrm{k}}} \tag{B.4.4}$$

and $\hat{\lambda}_i = 0 \ \forall i \notin \mathcal{W}_{\mathrm{k}}$. If $\hat{\lambda}_i \geq 0 \ \forall i \in \mathcal{W}_{\mathrm{k}}$, all the optimality conditions are satisfied so $\mathbf{x}_{k+1} = \mathbf{x}^\star$. If a multiplier $\hat{\lambda}_j, \ j \in \mathcal{W}_{\mathrm{k}}$ is negative, q can be decreased by removing j from $\mathcal{W}_{\mathrm{k}}$. We set $\mathcal{W}_{k+1} = \mathcal{W}_{\mathrm{k}} \setminus \{\hat{j}\}$ and $\mathbf{x}_{k+1} = \mathbf{x}_k$ (if there are several multipliers in this case, we choose $\hat{j} = \arg\min_j \ \hat{\lambda}_j$). The new point is still feasible for the new working set.

B.4.2 Application to non-negative LASSO

($\mathcal{P}$) can be re-written as

$$\mathbf{x}^\star \in \underset{\mathbf{x}\geq \mathbf{0}_n}{\operatorname{argmin}} \left\{ \frac{1}{2}\mathbf{x}^{\mathrm{T}} \underbrace{\mathbf{A}^{\mathrm{T}}\mathbf{A}}_{\mathbf{Q}} \mathbf{x} + \mathbf{x}^{\mathrm{T}} \underbrace{(\lambda \mathbf{1} - \mathbf{A}^{\mathrm{T}}\mathbf{y})}_{\mathbf{c}} \right\} \tag{B.4.5}$$

As the constraints are $x_i \geq 0 = \mathbf{e}_i{}^{\mathrm{T}}\mathbf{x} \geq b_i$ where $\mathbf{e}_i$ is the i-th canonical vector, we have $\mathbf{a}_i = \mathbf{e}_i$ and $b_i = 0$ so $\{\mathbf{a}_i\}_i$ are linearly independent vectors.

The only costly step in the AS method is the resolution of (B.4.2). For the non-negative LASSO, things are getting simpler. The solution of (B.4.2) is

$$\mathbf{p}_i = 0,\ \forall i \in \mathcal{W}_{\mathrm{k}} \qquad \text{and} \qquad \mathbf{p}_i = (-\mathbf{Q}^{-1}\mathbf{c})_i,\ \forall i \notin \mathcal{W}_{\mathrm{k}} \tag{B.4.6}$$

We can notice that a matrix inversion is required. However, we only need the columns and lines $i \notin \mathcal{W}_{\mathrm{k}}$ of $\mathbf{Q}$. Thus, we can use rank-one updates rules [Kha08] every time an element is added or removed from $\mathcal{W}_{\mathrm{k}}$. These rules cost $o(n^2)$ for a $n \times n$ matrix instead of $o(n^3)$ for a classical inversion [Kha08].

The initial inverse of $\mathbf{Q}$ can still be costly to compute if $\mathcal{W}_0$ is not well chosen. Indeed, the first inversion of $\mathbf{Q}$ should be computed anyway and costs $o(|\overline{\mathcal{W}}_0|^3)$ because no rank-one rules can be used. As we need to choose $\mathcal{W}_0$ as a subset of the active constraints of $\mathbf{x}_0$, we can choose all the active constraints of $\mathbf{x}_0$ is order to make $|\overline{\mathcal{W}}_0|$ as small as possible. Thus, a good starting point for (B.4.5) is $\mathbf{x}_0 = \mathbf{0}_n$.

C Proofs

C.1 Proof of (2.2.2)

Let $\mathbf{x}_\mathrm{f}$ a primal feasible point,

$$P(\mathbf{x}_\mathrm{f}) \geq P(\mathbf{x}^\star) = \frac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}^\star\|_2^2 + \lambda \mathbf{1}_n^\mathrm{T}\mathbf{x}^\star \geq \lambda \|\mathbf{x}^\star\|_1 \geq \lambda \|\mathbf{x}^\star\|_2 = \lambda \|\mathbf{x}^\star - \mathbf{0}\|_2 \tag{C.1.1}$$

C.2 Proof of (2.2.4)

Let $\mathbf{x}_\mathrm{f}$ a primal feasible point. The first-order optimality conditions for ($\mathcal{P}$) writes

$$\langle -\mathbf{A}^\mathrm{T}(\mathbf{y} - \mathbf{A}\mathbf{x}^\star) + \lambda \mathbf{1}, \mathbf{x}_\mathrm{f} - \mathbf{x}^\star \rangle \geq 0 \tag{C.2.1}$$

One also has

$$\begin{aligned}\frac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 &= \frac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}^\star + \mathbf{A}\mathbf{x}^\star - \mathbf{A}\mathbf{x}\|_2^2 \\ &= \frac{1}{2}\|\mathbf{y} - \mathbf{A}\mathbf{x}^\star\|_2^2 + \frac{1}{2}\|\mathbf{A}\mathbf{x}^\star - \mathbf{A}\mathbf{x}\|_2^2 + \langle -\mathbf{A}^\mathrm{T}(\mathbf{y} - \mathbf{A}\mathbf{x}^\star), \mathbf{x} - \mathbf{x}^\star \rangle\end{aligned} \tag{C.2.2}$$

Using (C.2.1) and (C.2.2), One can deduce that

$$\mathrm{P}(\mathbf{x}_\mathrm{f}) - \mathrm{P}(\mathbf{x}^\star) \geq \frac{1}{2}\|\mathbf{A}\mathbf{x}_\mathrm{f} - \mathbf{A}\mathbf{x}^\star\|_2^2 = \frac{1}{2}(\mathbf{x}^\star - \mathbf{x}_\mathrm{f})^\mathrm{T}\mathbf{A}^\mathrm{T}\mathbf{A}(\mathbf{x}^\star - \mathbf{x}_\mathrm{f})$$

One can finally use the weak duality $\mathrm{P}(\mathbf{x}^\star) \geq \mathrm{D}(\mathbf{u}^\star) \geq \mathrm{D}(\mathbf{u}_\mathrm{f})$ to conclude.

C.3 Proof of (2.2.3)

When $n \leq m$, $\mathbf{A}^\mathrm{T}\mathbf{A} - \alpha\mathbb{I}_n$ is positive definite as soon as $0 < \alpha \leq \lambda_{\min}(\mathbf{A}^\mathrm{T}\mathbf{A})$. As $\nabla^2\mathrm{P}(\mathbf{x}) = \mathbf{A}^\mathrm{T}\mathbf{A}$, we can deduce that P is α-strongly convex [Nes13]. Thus, $\forall\, \mathbf{x}_1, \mathbf{x}_2$, one has

$$\mathrm{P}(\mathbf{x}_1) \geq \mathrm{P}(\mathbf{x}_2) + \nabla\mathrm{P}(\mathbf{x}_2)^\mathrm{T}(\mathbf{x}_1 - \mathbf{x}_2) + \frac{\alpha}{2}\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 \tag{C.3.1}$$

Taking $\mathbf{x}_1 = \mathbf{x}_\mathrm{f}$ and $\mathbf{x}_2 = \mathbf{x}^\star$ and considering that $\mathrm{P}(\mathbf{x}^\star)^\mathrm{T}(\mathbf{x}_\mathrm{f} - \mathbf{x}^\star) \geq 0$ and $\mathrm{P}(\mathbf{x}^\star) \geq \mathrm{D}(\mathbf{u}_\mathrm{f})$ for any dual feasible point $\mathbf{u}_\mathrm{f}$ leads to the conclusion. This proof could also have been made by constructing the smallest sphere bounding (2.2.4), which is (2.2.3), using eigenvalue and semi-axis correspondence for ellipsoids.

C.4 Proof of (2.3.1)

Let $\tilde{\alpha} > 0$ a lower bound on $\lambda_{\max}(\mathbf{A}^\mathrm{T}\mathbf{A})$, let c the parameter allowing to tune the safeness of the dual-gap region and let $\kappa(\mathbf{X})$ the condition number of $\mathbf{X}$. As $\mathbf{A}^\mathrm{T}\mathbf{A}$ is normal and as $n \leq m$ when dual screening tests are used, we have

$\kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}{\lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}$. Furthermore, $\tilde{\alpha}$ is an upper bound on $\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})$. Thus, if $\kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A}) \leq c$,

$$0 < \frac{\tilde{\alpha}}{c} \leq \frac{\lambda_{\max}(\mathbf{A}^{\mathrm{T}}\mathbf{A})}{c} = \lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A})\frac{\kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A})}{c} \leq \lambda_{\min}(\mathbf{A}^{\mathrm{T}}\mathbf{A}) \tag{C.4.1}$$

So setting $\alpha = \frac{\tilde{\alpha}}{c}$ with $\kappa(\mathbf{A}^{\mathrm{T}}\mathbf{A}) \leq c$ creates a safe dual screening region.

References

[Bel65] Howard E Bell. "Gershgorin's theorem and the zeros of polynomials". In: *The American Mathematical Monthly* 72.3 (1965), pp. 292–295.

[BGT81] Robert G Bland, Donald Goldfarb, and Michael J Todd. "The ellipsoid method: A survey". In: *Operations research* 29.6 (1981), pp. 1039–1091.

[BHM98] Dimitri P Bertsekas, WW Hager, and OL Mangasarian. *Nonlinear programming*. Athena Scientific Belmont, MA, 1998.

[WN99] Stephen Wright and Jorge Nocedal. "Numerical optimization". In: *Springer Science* 35.67-68 (1999), p. 7.

[Hwa04] Suk-Geun Hwang. "Cauchy's interlace theorem for eigenvalues of Hermitian matrices". In: *The American Mathematical Monthly* 111.2 (2004), pp. 157–159.

[NW06] Jorge Nocedal and Stephen Wright. *Numerical optimization*. Springer Science & Business Media, 2006.

[Kha08] M Khan. *Updating inverse of a matrix when a column is added*. Tech. rep. removed. Technical report, University of British Columbia, Vancouver, 2008.

[BT09] Amir Beck and Marc Teboulle. "A fast iterative shrinkage-thresholding algorithm for linear inverse problems". In: *SIAM journal on imaging sciences* 2.1 (2009), pp. 183–202.

[GVR10] Laurent El Ghaoui, Vivian Viallon, and Tarek Rabbani. "Safe feature elimination for the lasso and sparse supervised learning problems". In: *arXiv preprint arXiv:1009.4219* (2010).

[BPC11] Stephen Boyd, Neal Parikh, and Eric Chu. *Distributed optimization and statistical learning via the alternating direction method of multipliers*. Now Publishers Inc, 2011.

[Jag13] Martin Jaggi. "Revisiting Frank-Wolfe: Projection-free sparse convex optimization". In: *Proceedings of the 30th international conference on machine learning*. CONF. 2013, pp. 427–435.

[Nes13] Yurii Nesterov. *Introductory lectures on convex optimization: A basic course*. Vol. 87. Springer Science & Business Media, 2013, pp. 75–76.

[Wan+13] Jie Wang et al. "Lasso screening rules via dual polytope projection". In: *Advances in neural information processing systems*. 2013, pp. 1070–1078.

[Bon+14] Antoine Bonnefoy et al. "A dynamic screening principle for the lasso". In: *2014 22nd European Signal Processing Conference (EUSIPCO)*. IEEE. 2014, pp. 6–10.

[MH16] Abed Malti and Cédric Herzet. "Safe screening tests for LASSO based on firmly non-expansiveness". In: *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE. 2016, pp. 4732–4736.

[XWR16] Zhen James Xiang, Yun Wang, and Peter J Ramadge. "Screening tests for lasso problems". In: *IEEE transactions on pattern analysis and machine intelligence* 39.5 (2016), pp. 1008–1027.

[Ndi+17] Eugene Ndiaye et al. "Gap safe screening rules for sparsity enforcing penalties". In: *The Journal of Machine Learning Research* 18.1 (2017), pp. 4671–4703.

[HDD19] Cedric Herzet, Clément Dorffer, and Angélique Drémeau. "Gather and conquer: Region-based strategies to accelerate safe screening tests". In: *IEEE Transactions on Signal Processing* 67.12 (2019), pp. 3300–3315.

[MdM19] Grégoire Mialon, Alexandre d'Aspremont, and Julien Mairal. "Screening Data Points in Empirical Risk Minimization via Ellipsoidal Regions and Safe Loss Function". In: *arXiv preprint arXiv:1912.02566* (2019).